

Standard Costs and the Balanced Scorecard

Chapter 10



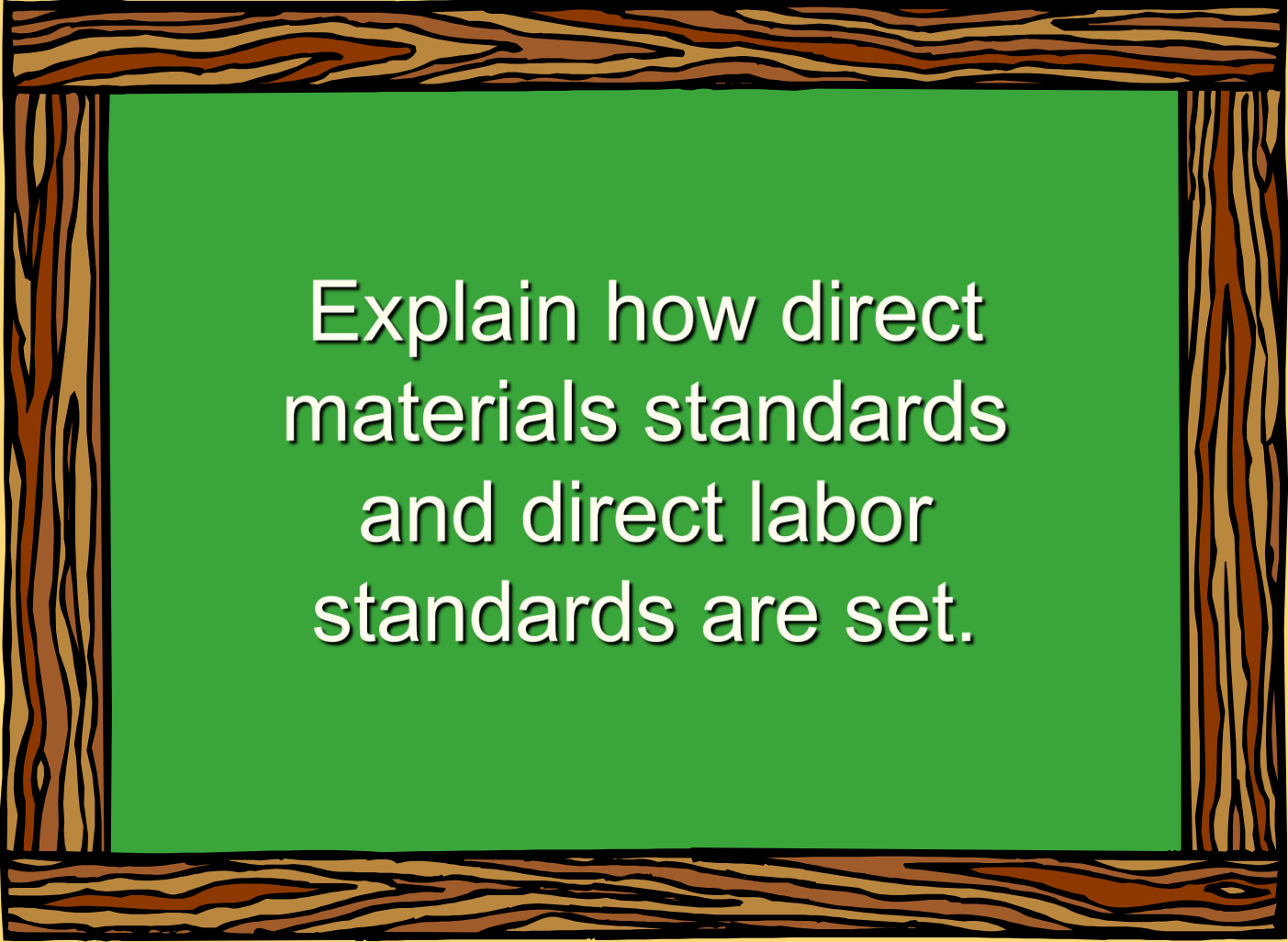
Standard Costs

Standards are benchmarks or “norms” for measuring performance. Two types of standards are commonly used.

Quantity standards specify how much of an input should be used to make a product or provide a service.

Cost (price) standards specify how much should be paid for each unit of the input.

Learning Objective 1

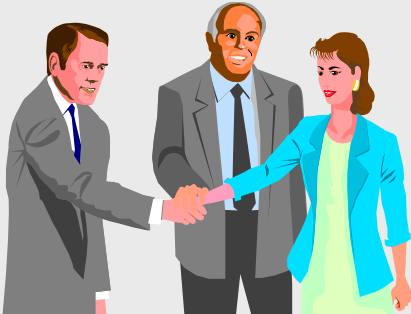


Explain how direct materials standards and direct labor standards are set.

Setting Direct Material Standards

Price Standards

**Final, delivered
cost of materials,
net of discounts.**



Quantity Standards

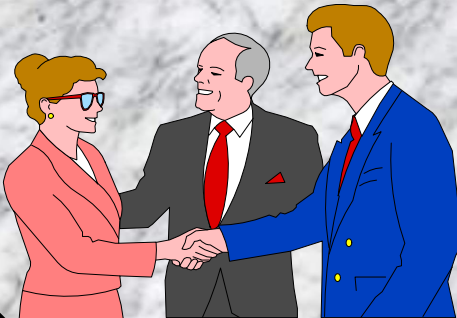
**Summarized in
a Bill of Materials.**



Setting Direct Labor Standards

Rate Standards

Often a single rate is used that reflects the mix of wages earned.



Time Standards

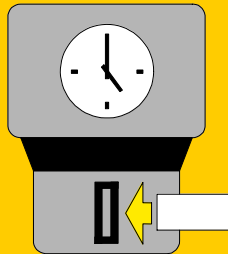
Use time and motion studies for each labor operation.



Setting Variable Overhead Standards

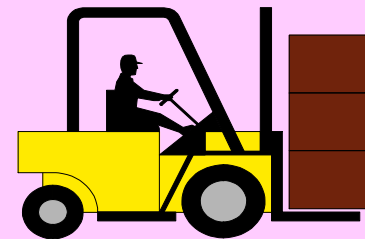
Rate Standards

The rate is the variable portion of the predetermined overhead rate.



Activity Standards

The activity is the base used to calculate the predetermined overhead.

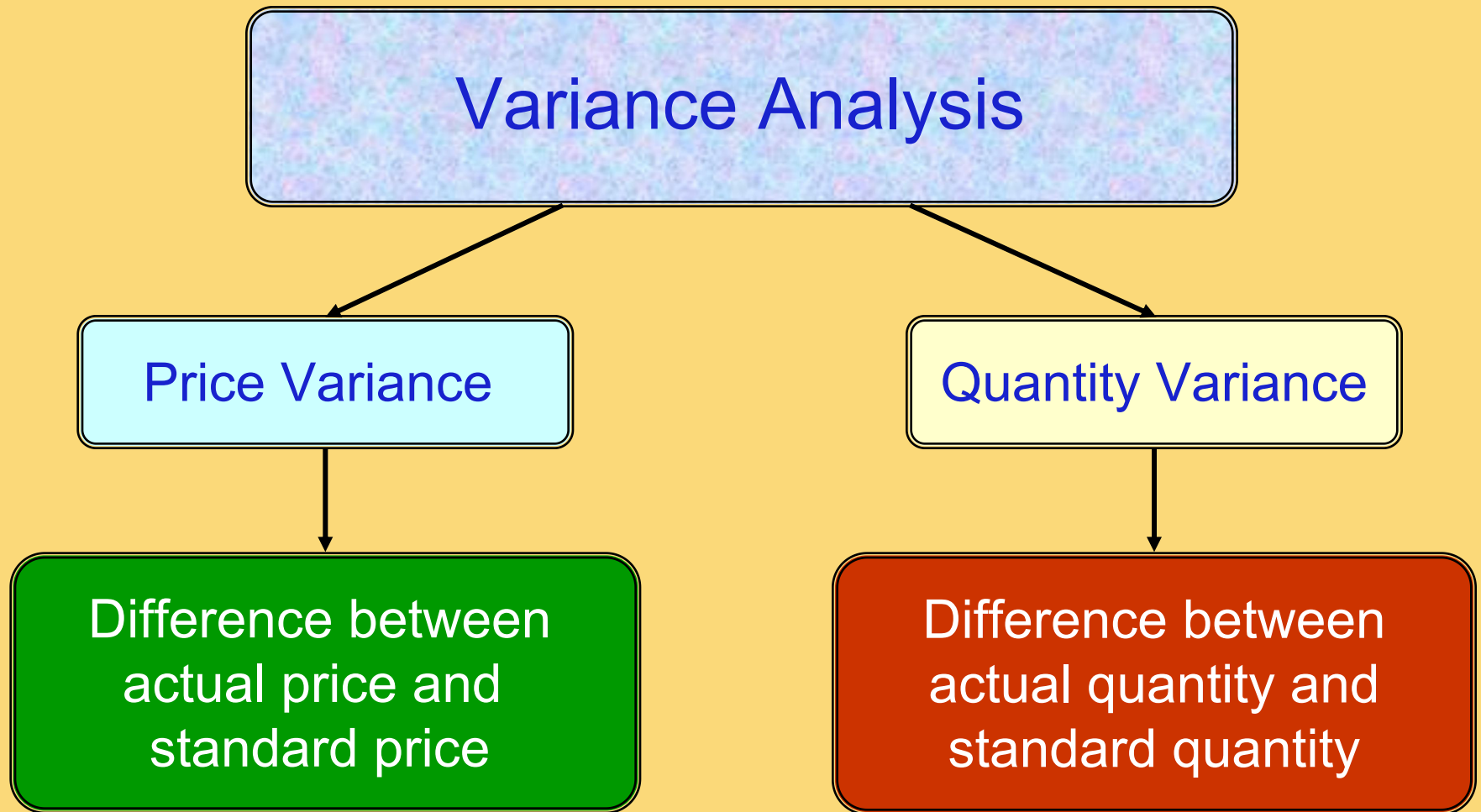


Standard Cost Card – Variable Production Cost

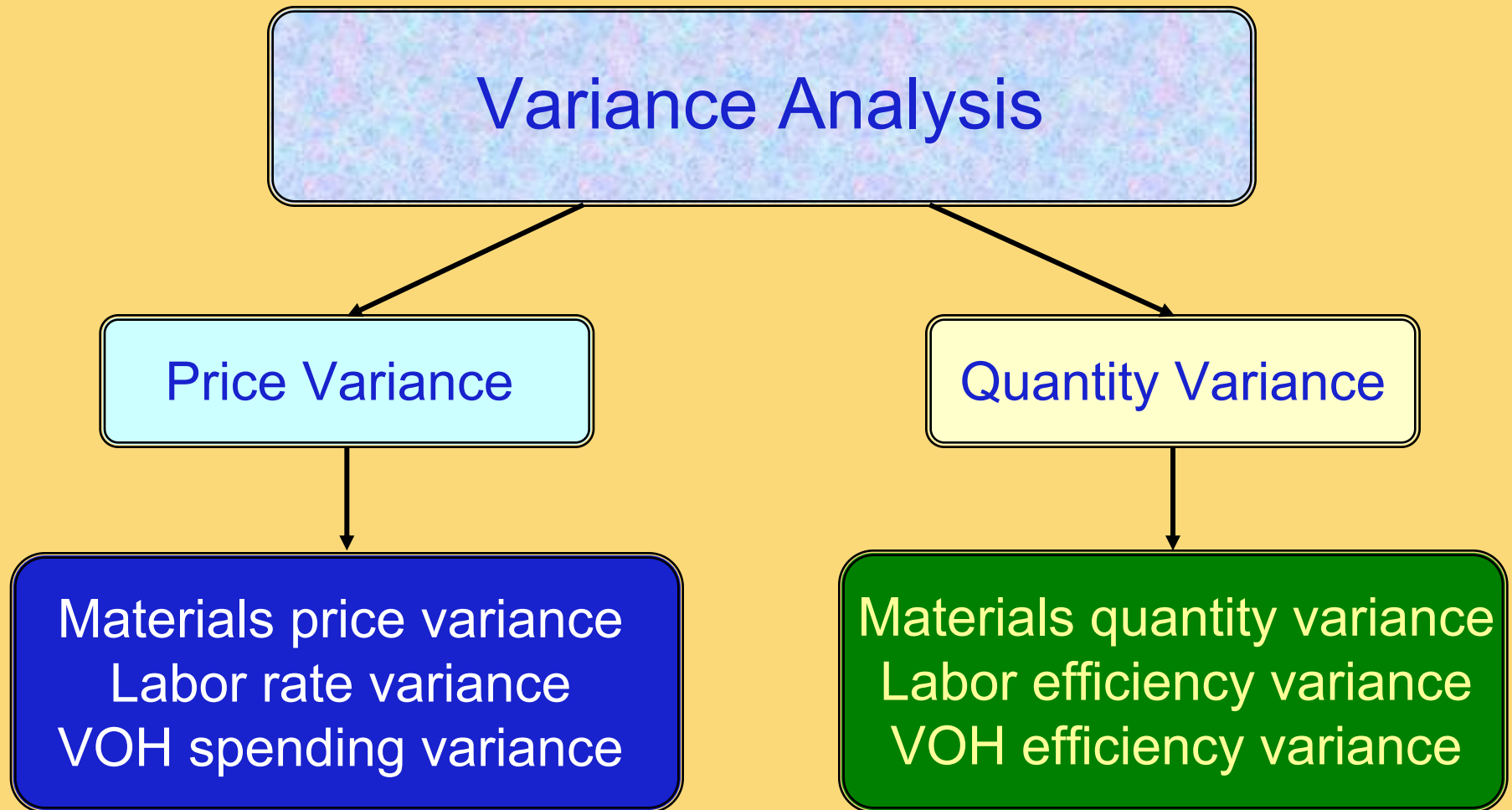
A standard cost card for one unit of product might look like this:

Inputs	A	B	A x B
	Standard Quantity or Hours	Standard Price or Rate	Standard Cost per Unit
Direct materials	3.0 lbs.	\$ 4.00 per lb.	\$ 12.00
Direct labor	2.5 hours	14.00 per hour	35.00
Variable mfg. overhead	2.5 hours	3.00 per hour	7.50
Total standard unit cost			\$ 54.50

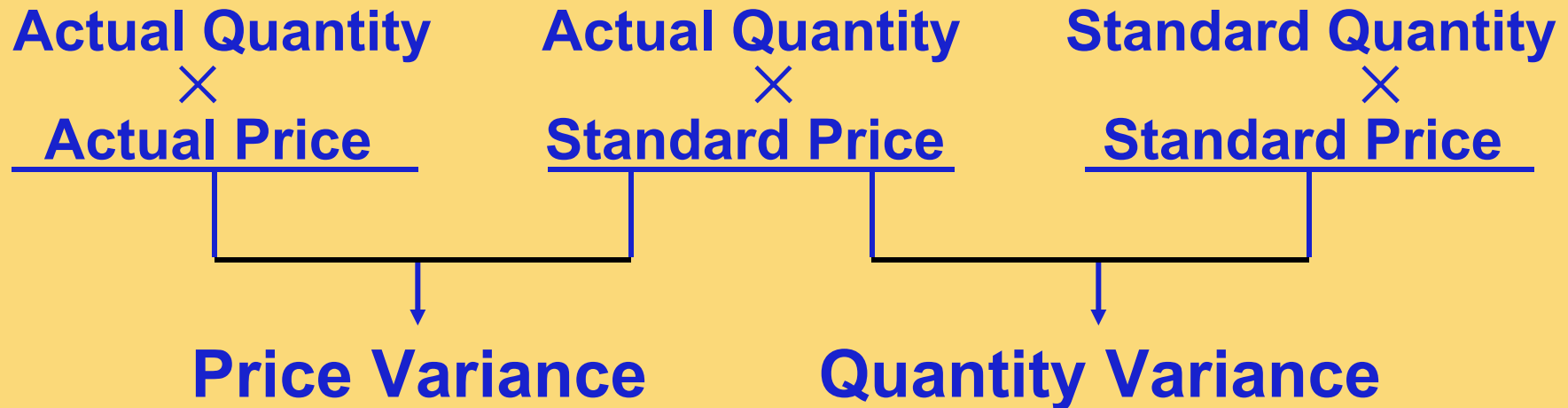
A General Model for Variance Analysis



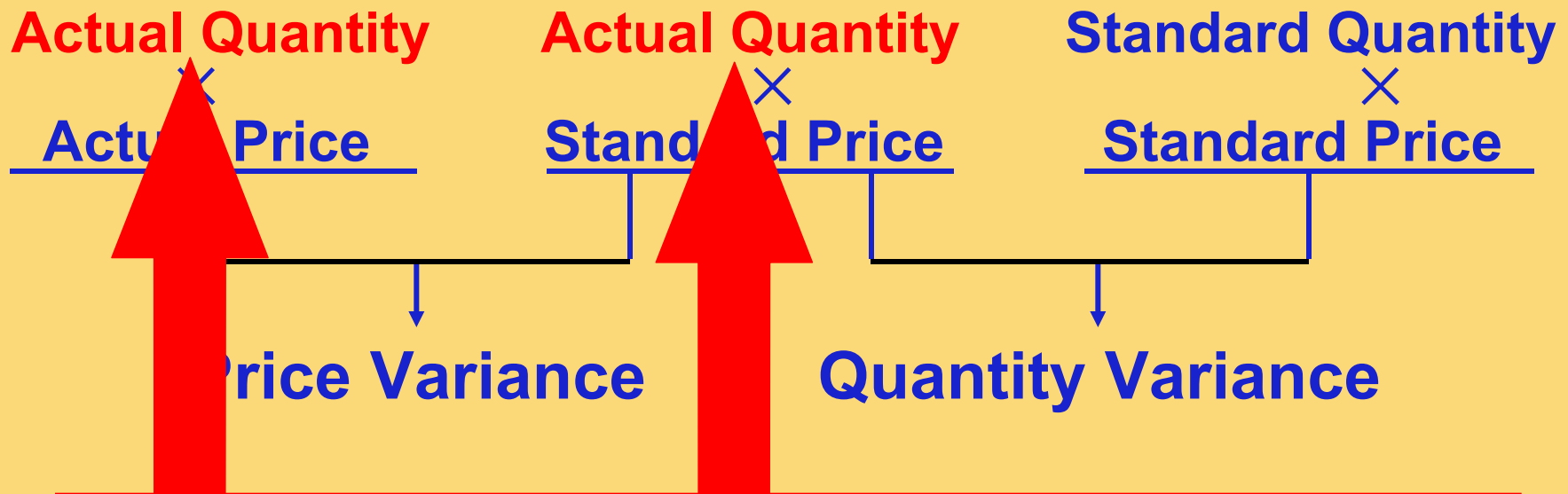
A General Model for Variance Analysis



A General Model for Variance Analysis

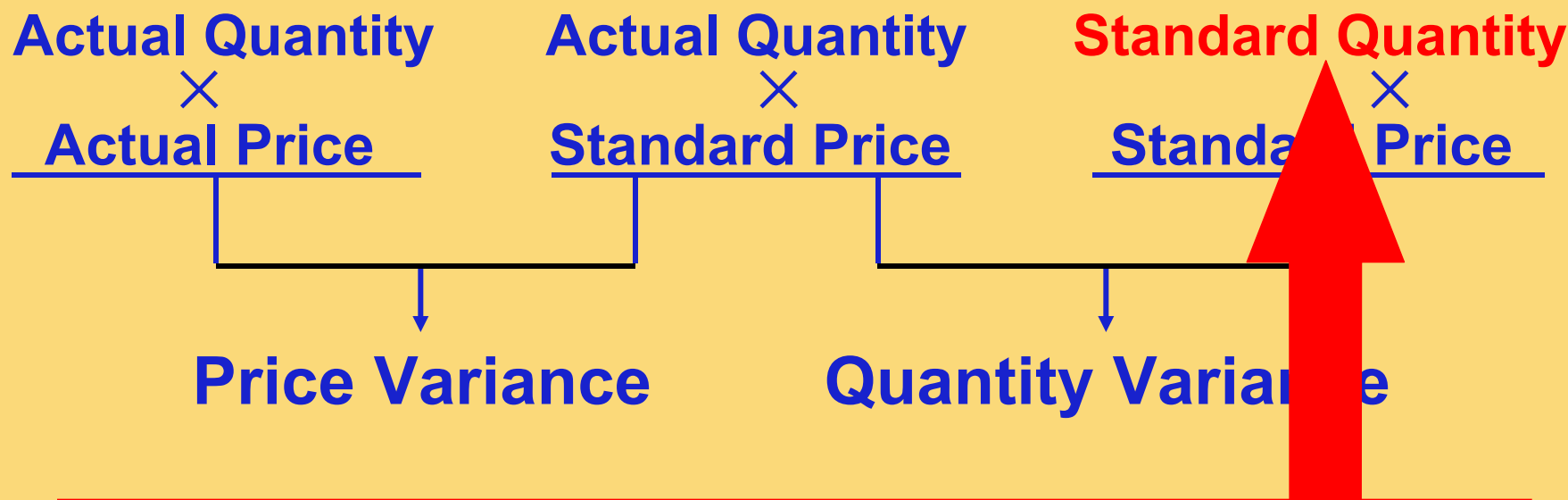


A General Model for Variance Analysis



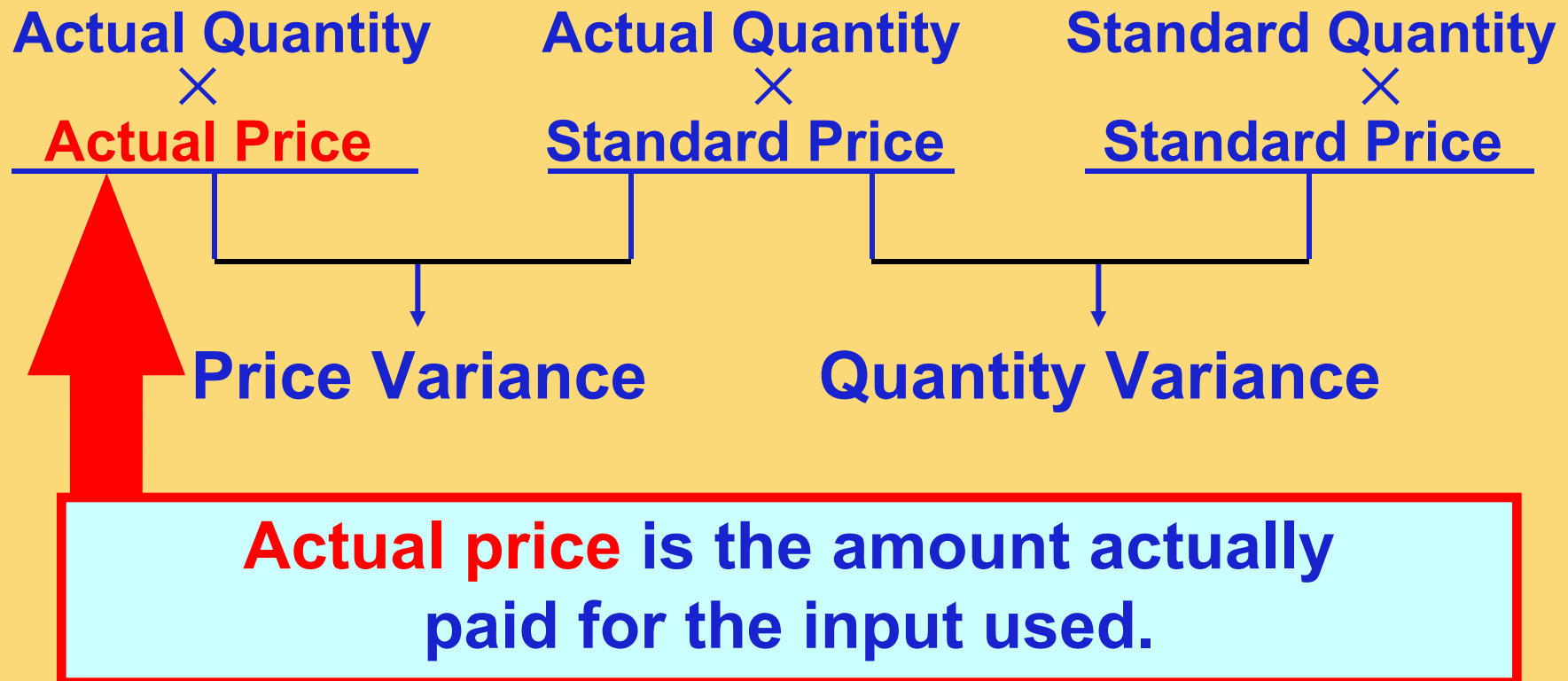
Actual quantity is the amount of direct materials, direct labor, and variable manufacturing overhead actually used.

A General Model for Variance Analysis

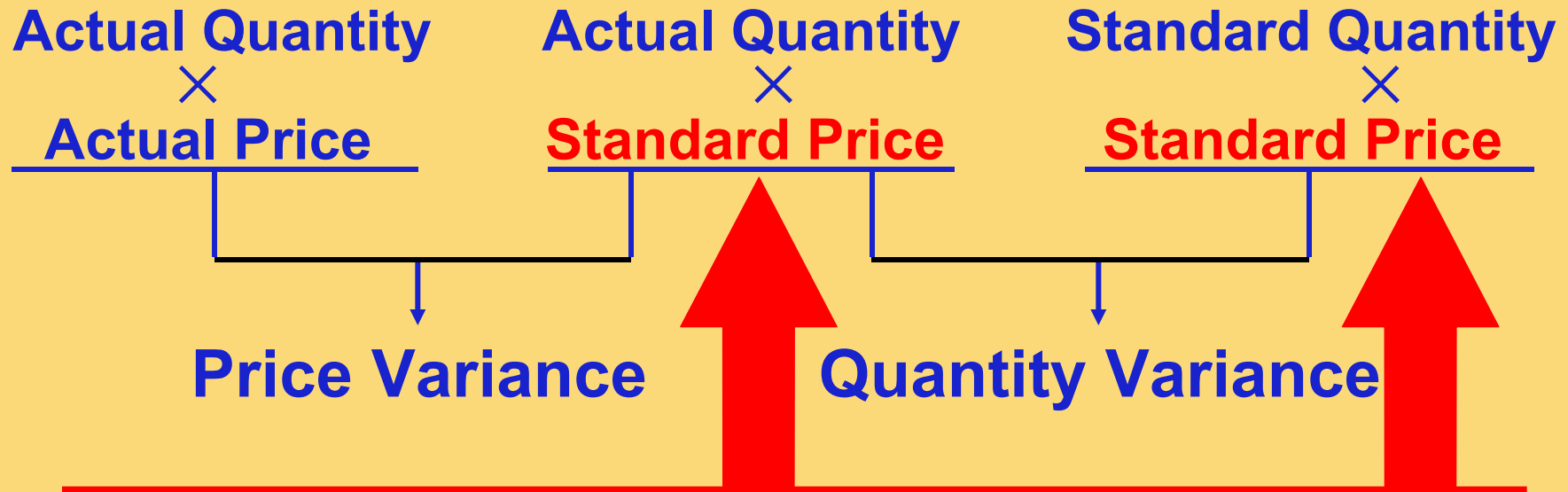


Standard quantity is the standard quantity allowed for the actual output of the period.

A General Model for Variance Analysis

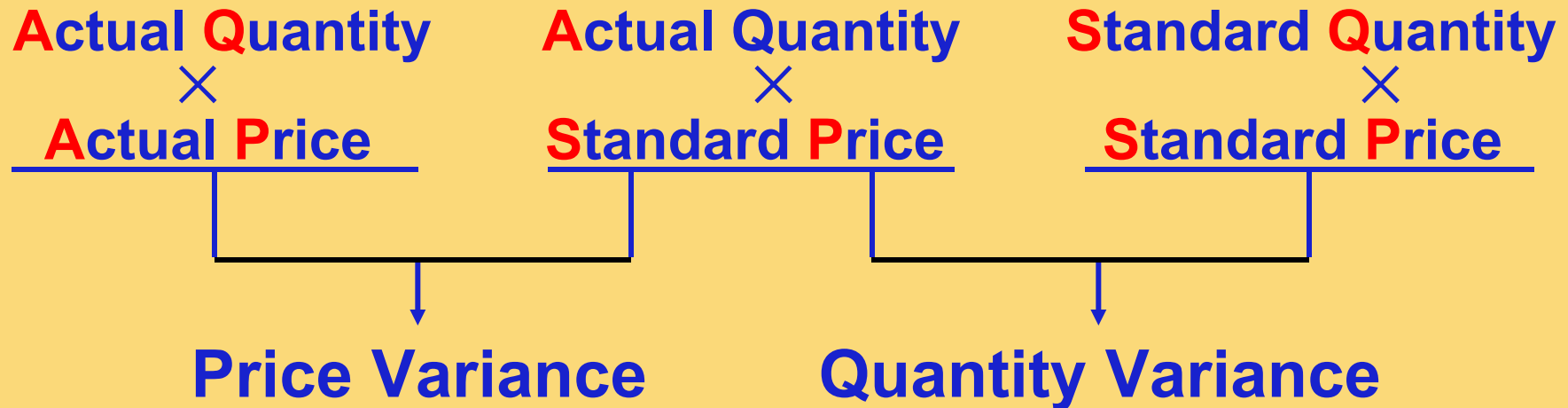


A General Model for Variance Analysis



Standard price is the amount that should have been paid for the input used.

A General Model for Variance Analysis



$$(AQ \times AP) - (AQ \times SP)$$

AQ = Actual Quantity

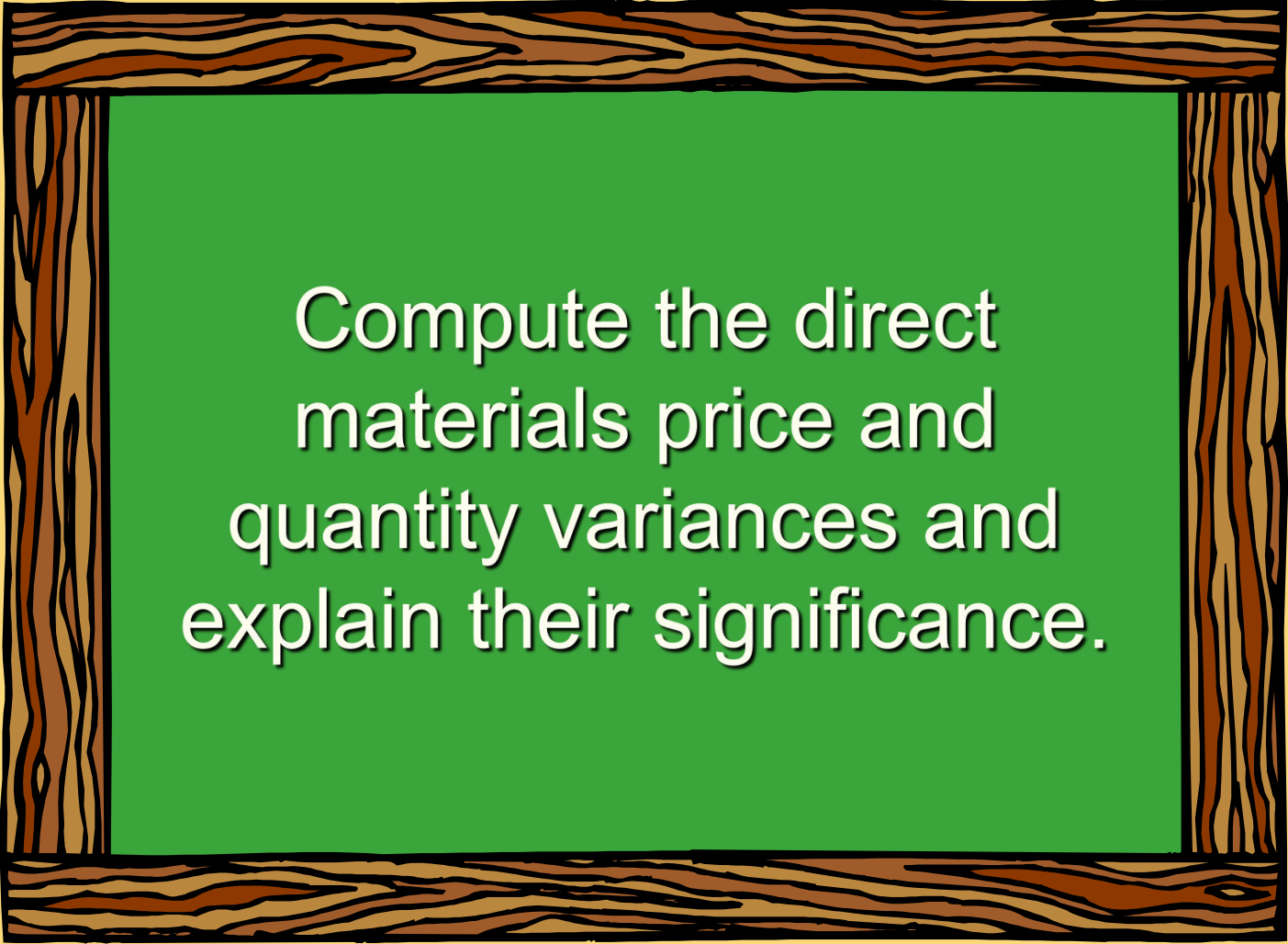
AP = Actual Price

$$(AQ \times SP) - (SQ \times SP)$$

SP = Standard Price

SQ = Standard Quantity

Learning Objective 2



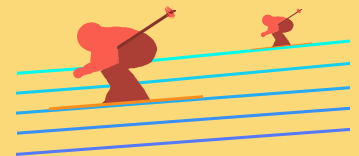
Compute the direct materials price and quantity variances and explain their significance.

Material Variances Example

Glacier Peak Outfitters has the following direct material standard for the fiberfill in its mountain parka.

0.1 kg. of fiberfill per parka at \$5.00 per kg.

Last month 210 kgs of fiberfill were purchased and used to make 2,000 parkas. The material cost a total of \$1,029.



Material Variances Summary

Actual Quantity
×
Actual Price

210 kgs.

×

\$4.90 per kg.

= \$1,029

Actual Quantity
×
Standard Price

210 kgs.

×

\$5.00 per kg.

= \$1,050

Standard Quantity
×
Standard Price

200 kgs.

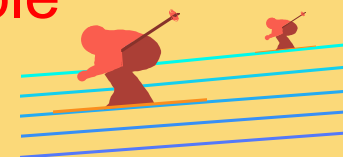
×

\$5.00 per kg.

= \$1,000

Price variance
\$21 favorable

Quantity variance
\$50 unfavorable



Material Variances Summary

Actual Quantity
×
Actual Price

210 kgs.

×

\$4.90 per kg.

= \$1,029

Actual Quantity
×
Standard Price

210 kgs.

×

\$1,029 ÷ 210 kgs
= \$4.90 per kg

= \$1,050

Standard Quantity
×
Standard Price

200 kgs.

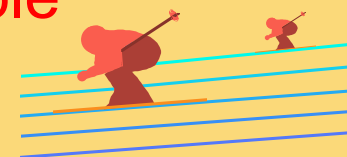
×

\$5.00 per kg.

= \$1,000

Price variance
\$21 favorable

Quantity variance
\$50 unfavorable



Material Variances Summary

Actual Quantity
×
Actual Price

210 kgs.

×

\$4.90 per

= \$1,029

Actual Quantity
×
Standard Price

210 kgs.

×

Standard Price

= \$1,050

Standard Quantity
×
Standard Price

200 kgs.

×

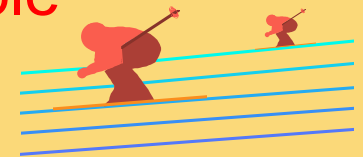
\$5.00 per kg.

= \$1,000

0.1 kg per parka × 2,000 parkas
= 200 kgs

Price variance
\$21 favorable

Quantity variance
\$50 unfavorable



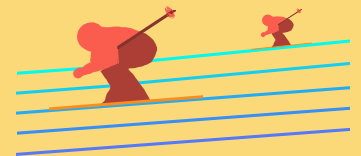
Material Variances: Using the Factored Equations

Materials price variance

$$\begin{aligned}\text{MPV} &= \text{AQ} (\text{AP} - \text{SP}) \\ &= 210 \text{ kgs} (\$4.90/\text{kg} - \$5.00/\text{kg}) \\ &= 210 \text{ kgs} (-\$0.10/\text{kg}) \\ &= \$21 \text{ F}\end{aligned}$$

Materials quantity variance

$$\begin{aligned}\text{MQV} &= \text{SP} (\text{AQ} - \text{SQ}) \\ &= \$5.00/\text{kg} (210 \text{ kgs} - (0.1 \text{ kg/parka} \times 2,000 \text{ parkas})) \\ &= \$5.00/\text{kg} (210 \text{ kgs} - 200 \text{ kgs}) \\ &= \$5.00/\text{kg} (10 \text{ kgs}) \\ &= \$50 \text{ U}\end{aligned}$$



Isolation of Material Variances

I need the price variance sooner so that I can better identify purchasing problems.

You accountants just don't understand the problems that **purchasing managers** have.



I'll start computing the price variance when material is purchased rather than when it's used.



Material Variances



Hanson purchased and used 1,700 pounds. How are the variances computed if the amount purchased **differs** from the amount used?



The price variance is computed on the entire quantity **purchased**.

The quantity variance is computed only on the quantity **used**.

Responsibility for Material Variances

Materials Quantity Variance



Production Manager

Materials Price Variance



Purchasing Manager

The standard price is used to compute the quantity variance so that the production manager is not held responsible for the purchasing manager's performance.

Responsibility for Material Variances

I am not responsible for this unfavorable material quantity variance.

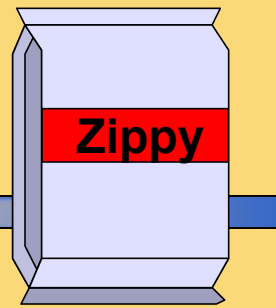
You purchased cheap material, so my people had to use more of it.



Your poor scheduling sometimes requires me to rush order material at a higher price, causing unfavorable price variances.



Quick Check ✓

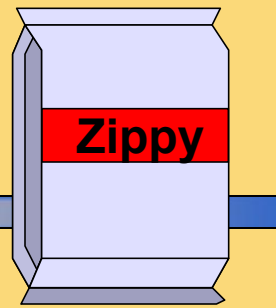


Hanson Inc. has the following direct material standard to manufacture one Zippy:

1.5 pounds per Zippy at \$4.00 per pound

Last week, 1,700 pounds of material were purchased and used to make 1,000 Zippies.
The material cost a total of \$6,630.

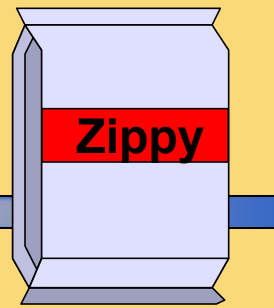
Quick Check ✓



Hanson's material price variance (MPV) for the week was:

- a. \$170 unfavorable.
- b. \$170 favorable.
- c. \$800 unfavorable.
- d. \$800 favorable.

Quick Check ✓



Hanson's material price variance (MPV) for the week was:

a. \$170 unfavorable.

b. \$170 favorable.

c. \$800 unfavorable.

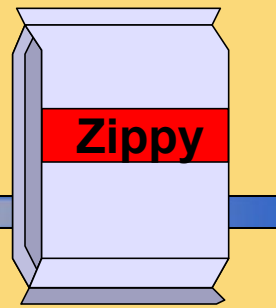
d. \$800 favorable.

$$\text{MPV} = \text{AQ}(\text{AP} - \text{SP})$$

$$\text{MPV} = 1,700 \text{ lbs.} \times (\$3.90 - 4.00)$$

$$\text{MPV} = \$170 \text{ Favorable}$$

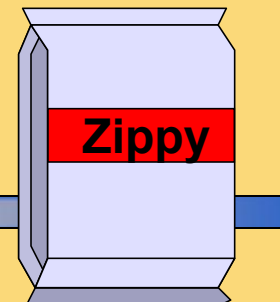
Quick Check ✓



Hanson's material quantity variance (MQV) for the week was:

- a. \$170 unfavorable.
- b. \$170 favorable.
- c. \$800 unfavorable.
- d. \$800 favorable.

Quick Check ✓



Hanson's material quantity variance (MQV) for the week was:

a. \$170 unfavorable.

b. \$170 favorable.

c. \$800 unfavorable.

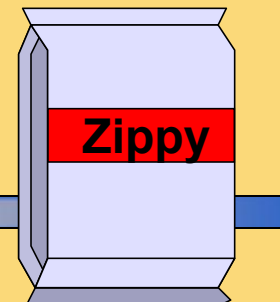
d. \$800 favorable.

$$\text{MQV} = \text{SP}(\text{AQ} - \text{SQ})$$

$$\text{MQV} = \$4.00(1,700 \text{ lbs} - 1,500 \text{ lbs})$$

$$\text{MQV} = \$800 \text{ unfavorable}$$

Quick Check ✓



Actual Quantity
×
Actual Price

1,700 lbs.

×

\$3.90 per lb.

= \$6,630

Actual Quantity
×
Standard Price

1,700 lbs.

×

\$4.00 per lb.

= \$ 6,800

Standard Quantity
×
Standard Price

1,500 lbs.

×

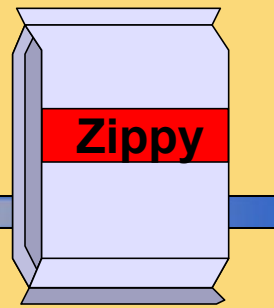
\$4.00 per lb.

= \$6,000

Price variance
\$170 favorable

Quantity variance
\$800 unfavorable

Quick Check ✓ Continued

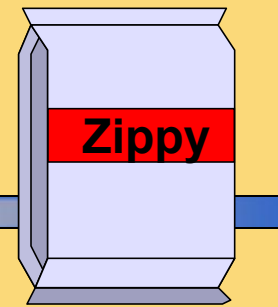


Hanson Inc. has the following material standard to manufacture one Zippy:

1.5 pounds per Zippy at \$4.00 per pound

Last week, **2,800 pounds** of material were purchased at a total cost of \$10,920, and 1,700 pounds were used to make 1,000 Zippies.

Quick Check ✓ Continued



**Actual Quantity
Purchased**
×
Actual Price

2,800 lbs.

×

\$3.90 per lb.

= \$10,920

**Actual Quantity
Purchased**
×
Standard Price

2,800 lbs.

×

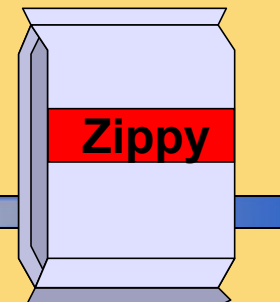
\$4.00 per lb.

= \$11,200

Price variance
\$280 favorable

Price variance increases
because quantity
purchased increases.

Quick Check ✓ Continued



Actual Quantity
Used
×
Standard Price

1,700 lbs.

×

\$4.00 per lb.

= \$6,800

Standard Quantity
×
Standard Price

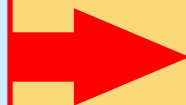
1,500 lbs.

×

\$4.00 per lb.

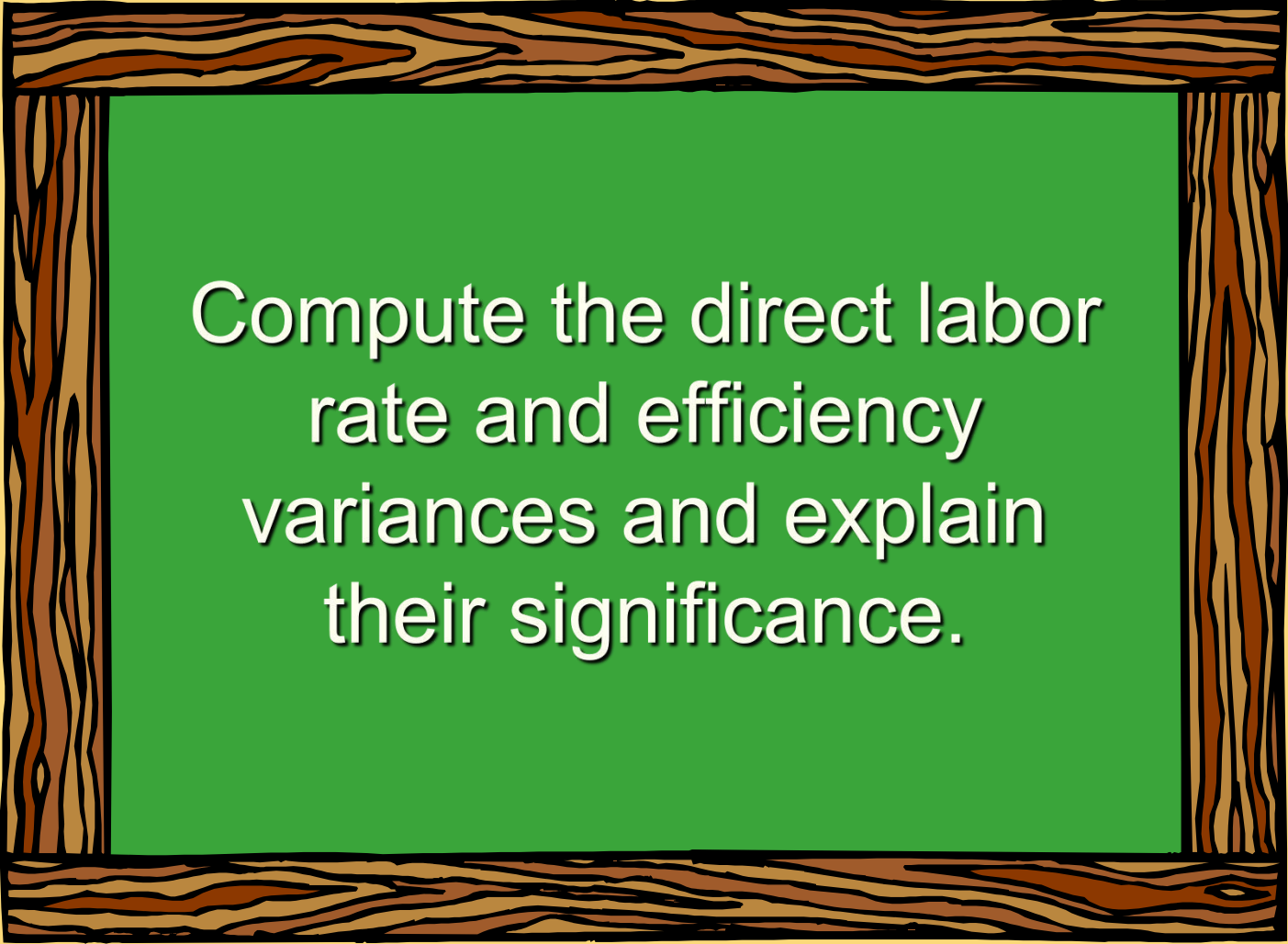
= \$6,000

Quantity variance is unchanged because actual and standard quantities are unchanged.



Quantity variance
\$800 unfavorable

Learning Objective 3



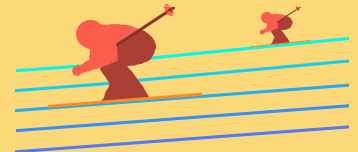
Compute the direct labor
rate and efficiency
variances and explain
their significance.

Labor Variances Example

Glacier Peak Outfitters has the following direct labor standard for its mountain parka.

1.2 standard hours per parka at \$10.00 per hour

Last month, employees actually worked 2,500 hours at a total labor cost of \$26,250 to make 2,000 parkas.



Labor Variances Summary

$$\begin{array}{r} \text{Actual Hours} \\ \times \\ \hline \text{Actual Rate} \end{array}$$

2,500 hours

$$\begin{array}{r} \times \\ \$10.50 \text{ per hour} \end{array}$$

= \$26,250

$$\begin{array}{r} \text{Actual Hours} \\ \times \\ \hline \text{Standard Rate} \end{array}$$

2,500 hours

$$\begin{array}{r} \times \\ \$10.00 \text{ per hour.} \end{array}$$

= \$25,000

$$\begin{array}{r} \text{Standard Hours} \\ \times \\ \hline \text{Standard Rate} \end{array}$$

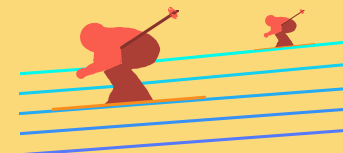
2,400 hours

$$\begin{array}{r} \times \\ \$10.00 \text{ per hour} \end{array}$$

= \$24,000

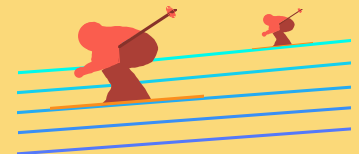
Rate variance
\$1,250 unfavorable

Efficiency variance
\$1,000 unfavorable



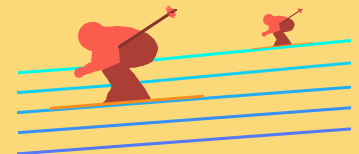
Labor Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours	2,500 hours	2,400 hours
×		×
\$10.50 per hour	$\\$26,250 \div 2,500 \text{ hours}$ $= \\$10.50 \text{ per hour}$	\$10.00 per hour
= \$26,250	= \$25,000	= \$24,000
Rate variance \$1,250 unfavorable		Efficiency variance \$1,000 unfavorable



Labor Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours	2,500 hours	2,400 hours
×		×
\$10.50 per hour	1.2 hours per parka × 2,000 parkas = 2,400 hours	10.00 per hour
= \$26,250	= \$25,000	= \$24,000
Rate variance \$1,250 unfavorable		Efficiency variance \$1,000 unfavorable



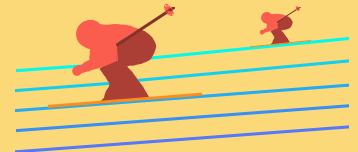
Labor Variances: Using the Factored Equations

Labor rate variance

$$\begin{aligned}\text{LRV} &= \text{AH} (\text{AR} - \text{SR}) \\ &= 2,500 \text{ hours } (\$10.50 \text{ per hour} - \$10.00 \text{ per hour}) \\ &= 2,500 \text{ hours } (\$0.50 \text{ per hour}) \\ &= \$1,250 \text{ unfavorable}\end{aligned}$$

Labor efficiency variance

$$\begin{aligned}\text{LEV} &= \text{SR} (\text{AH} - \text{SH}) \\ &= \$10.00 \text{ per hour } (2,500 \text{ hours} - 2,400 \text{ hours}) \\ &= \$10.00 \text{ per hour } (100 \text{ hours}) \\ &= \$1,000 \text{ unfavorable}\end{aligned}$$



Responsibility for Labor Variances

Production managers are usually held accountable for labor variances because they can influence the:



Production Manager

Mix of skill levels assigned to work tasks.

Level of employee motivation.

Quality of production supervision.

Quality of training provided to employees.

Responsibility for Labor Variances

I am not responsible for the unfavorable labor efficiency variance!

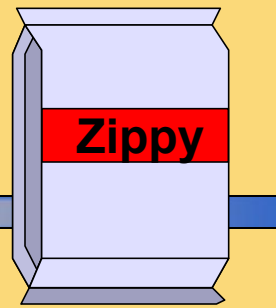
You purchased cheap material, so it took more time to process it.



I think it took more time to process the materials because the Maintenance Department has poorly maintained your equipment.



Quick Check ✓

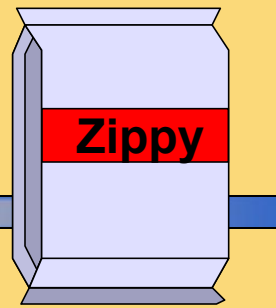


Hanson Inc. has the following direct labor standard to manufacture one Zippy:

1.5 standard hours per Zippy at \$12.00 per direct labor hour

Last week, 1,550 direct labor hours were worked at a total labor cost of \$18,910 to make 1,000 Zippies.

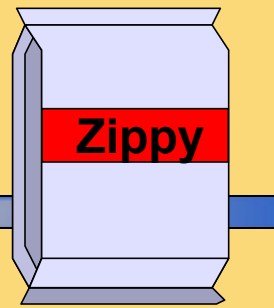
Quick Check ✓



Hanson's labor rate variance (LRV) for the week was:

- a. \$310 unfavorable.
- b. \$310 favorable.
- c. \$300 unfavorable.
- d. \$300 favorable.

Quick Check ✓



Hanson's labor rate variance (LRV) for the week was:

a. \$310 unfavorable.

b. \$310 favorable.

c. \$300 unfavorable.

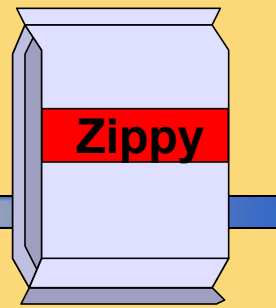
d. \$300 favorable.

$$\text{LRV} = \text{AH}(\text{AR} - \text{SR})$$

$$\text{LRV} = 1,550 \text{ hrs}(\$12.20 - \$12.00)$$

$$\text{LRV} = \$310 \text{ unfavorable}$$

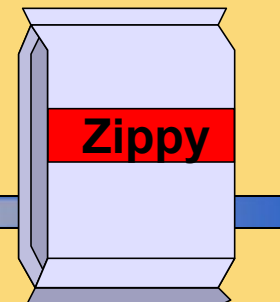
Quick Check ✓



Hanson's labor efficiency variance (LEV) for the week was:

- a. \$590 unfavorable.
- b. \$590 favorable.
- c. \$600 unfavorable.
- d. \$600 favorable.

Quick Check ✓



Hanson's labor efficiency variance (LEV) for the week was:

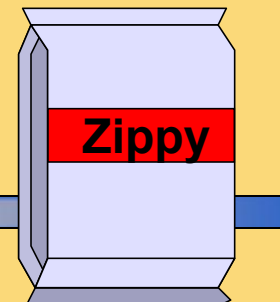
- a. \$590 unfavorable.
- b. \$590 favorable.
- c. \$600 unfavorable.**
- d. \$600 favorable.

$$\text{LEV} = \text{SR}(\text{AH} - \text{SH})$$

$$\text{LEV} = \$12.00(1,550 \text{ hrs} - 1,500 \text{ hrs})$$

$$\text{LEV} = \$600 \text{ unfavorable}$$

Quick Check ✓



Actual Hours
×
Actual Rate

1,550 hours

×

\$12.20 per hour

= \$18,910

Actual Hours
×
Standard Rate

1,550 hours

×

\$12.00 per hour

= \$18,600

Standard Hours
×
Standard Rate

1,500 hours

×

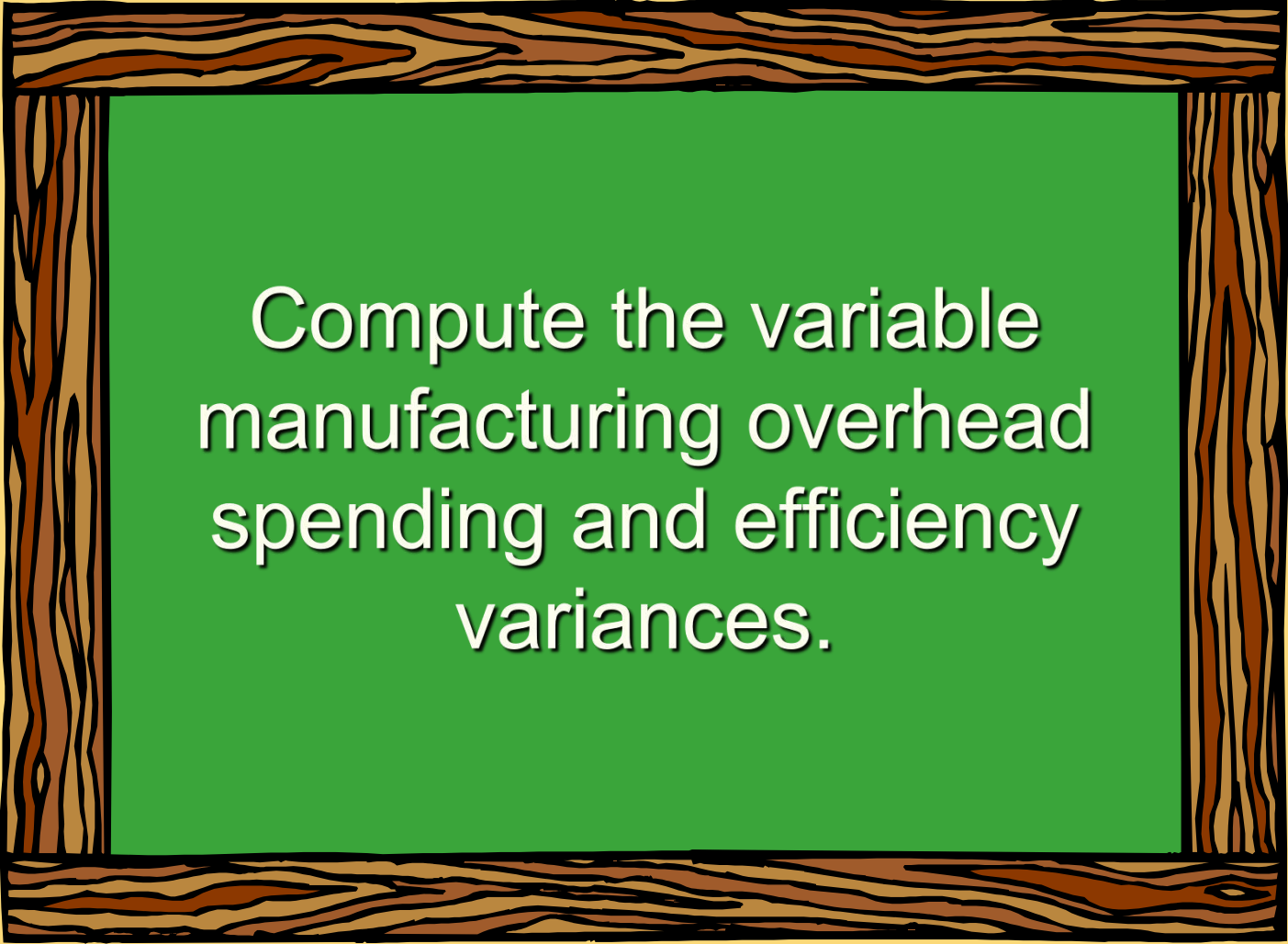
\$12.00 per hour

= \$18,000

Rate variance
\$310 unfavorable

Efficiency variance
\$600 unfavorable

Learning Objective 4



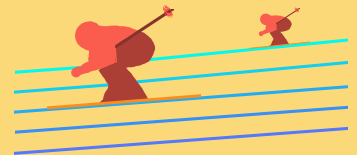
Compute the variable
manufacturing overhead
spending and efficiency
variances.

Variable Manufacturing Overhead Variances Example

Glacier Peak Outfitters has the following direct variable manufacturing overhead labor standard for its mountain parka.

1.2 standard hours per parka at \$4.00 per hour

Last month, employees actually worked 2,500 hours to make 2,000 parkas. Actual variable manufacturing overhead for the month was \$10,500.



Variable Manufacturing Overhead Variances Summary

Actual Hours
×
Actual Rate

2,500 hours

×

\$4.20 per hour

= \$10,500

Actual Hours
×
Standard Rate

2,500 hours

×

\$4.00 per hour

= \$10,000

Standard Hours
×
Standard Rate

2,400 hours

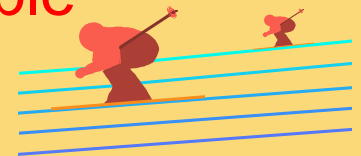
×

\$4.00 per hour

= \$9,600

Spending variance
\$500 unfavorable

Efficiency variance
\$400 unfavorable



Variable Manufacturing Overhead Variances Summary

Actual Hours
×
Actual Rate

2,500 hours

×
\$4.20 per hour

= \$10,500

Actual Hours
×
Standard Rate

2,500 hours

$\$10,500 \div 2,500 \text{ hours}$
= \$4.20 per hour

= \$10,000

Standard Hours
×
Standard Rate

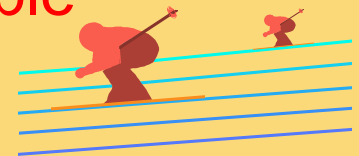
2,400 hours

×
\$4.00 per hour

= \$9,600

Spending variance
\$500 unfavorable

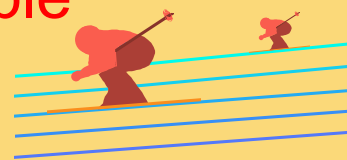
Efficiency variance
\$400 unfavorable



Variable Manufacturing Overhead Variances Summary

Actual Hours × Actual Rate	Actual Hours × Standard Rate	Standard Hours × Standard Rate
2,500 hours × \$4.20 per hour	2,500 hours × \$4.00 per hour	2,400 hours × \$4.00 per hour
= \$10,500	= \$10,000	= \$9,600
Spending variance \$500 unfavorable		Efficiency variance \$400 unfavorable

1.2 hours per parka × 2,000 parkas = 2,400 hours



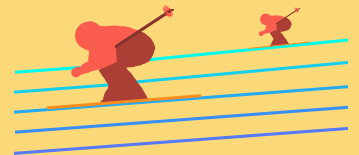
Variable Manufacturing Overhead Variances: Using Factored Equations

Variable manufacturing overhead spending variance

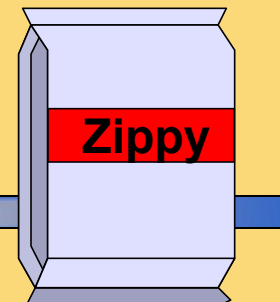
$$\begin{aligned}\text{VMSV} &= \text{AH} (\text{AR} - \text{SR}) \\ &= 2,500 \text{ hours } (\$4.20 \text{ per hour} - \$4.00 \text{ per hour}) \\ &= 2,500 \text{ hours } (\$0.20 \text{ per hour}) \\ &= \$500 \text{ unfavorable}\end{aligned}$$

Variable manufacturing overhead efficiency variance

$$\begin{aligned}\text{VMEV} &= \text{SR} (\text{AH} - \text{SH}) \\ &= \$4.00 \text{ per hour } (2,500 \text{ hours} - 2,400 \text{ hours}) \\ &= \$4.00 \text{ per hour } (100 \text{ hours}) \\ &= \$400 \text{ unfavorable}\end{aligned}$$



Quick Check ✓

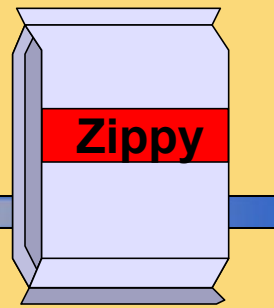


Hanson Inc. has the following variable manufacturing overhead standard to manufacture one Zippy:

1.5 standard hours per Zippy at \$3.00 per direct labor hour

Last week, 1,550 hours were worked to make 1,000 Zippies, and \$5,115 was spent for variable manufacturing overhead.

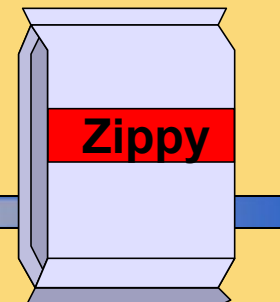
Quick Check ✓



Hanson's spending variance (VOSV) for variable manufacturing overhead for the week was:

- a. \$465 unfavorable.
- b. \$400 favorable.
- c. \$335 unfavorable.
- d. \$300 favorable.

Quick Check ✓



Hanson's spending variance (VOSV) for variable manufacturing overhead for the week was:

a. \$465 unfavorable.

b. \$400 favorable

c. \$335 unfavorable

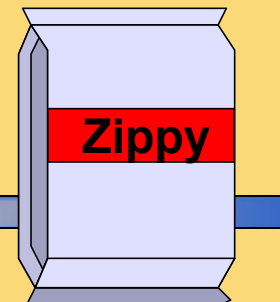
d. \$300 favorable

$$\text{VOSV} = \text{AH}(\text{AR} - \text{SR})$$

$$\text{VOSV} = 1,550 \text{ hrs}(\$3.30 - \$3.00)$$

$$\text{VOSV} = \$465 \text{ unfavorable}$$

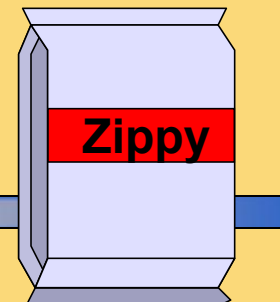
Quick Check ✓



Hanson's efficiency variance (VOEV) for variable manufacturing overhead for the week was:

- a. \$435 unfavorable.
- b. \$435 favorable.
- c. \$150 unfavorable.
- d. \$150 favorable.

Quick Check ✓



Hanson's efficiency variance (VOEV) for variable manufacturing overhead for the week was:

a. \$435 unfavorable.

b. \$435 favorable.

c. \$150 unfavorable

d. \$150 favorable

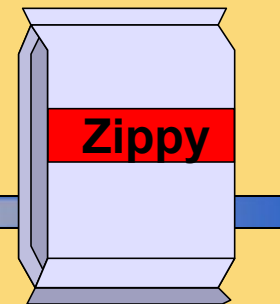
1,000 units × 1.5 hrs per unit

$$\text{VOEV} = \text{SR}(\text{AH} - \text{SH})$$

$$\text{VOEV} = \$3.00(1,550 \text{ hrs} - 1,500 \text{ hrs})$$

$$\text{VOEV} = \$150 \text{ unfavorable}$$

Quick Check ✓



Actual Hours
×
Actual Rate

1,550 hours

×

\$3.30 per hour

= \$5,115

Actual Hours
×
Standard Rate

1,550 hours

×

\$3.00 per hour

= \$4,650

Standard Hours
×
Standard Rate

1,500 hours

×

\$3.00 per hour

= \$4,500

Spending variance
\$465 unfavorable

Efficiency variance
\$150 unfavorable

Variance Analysis and Management by Exception

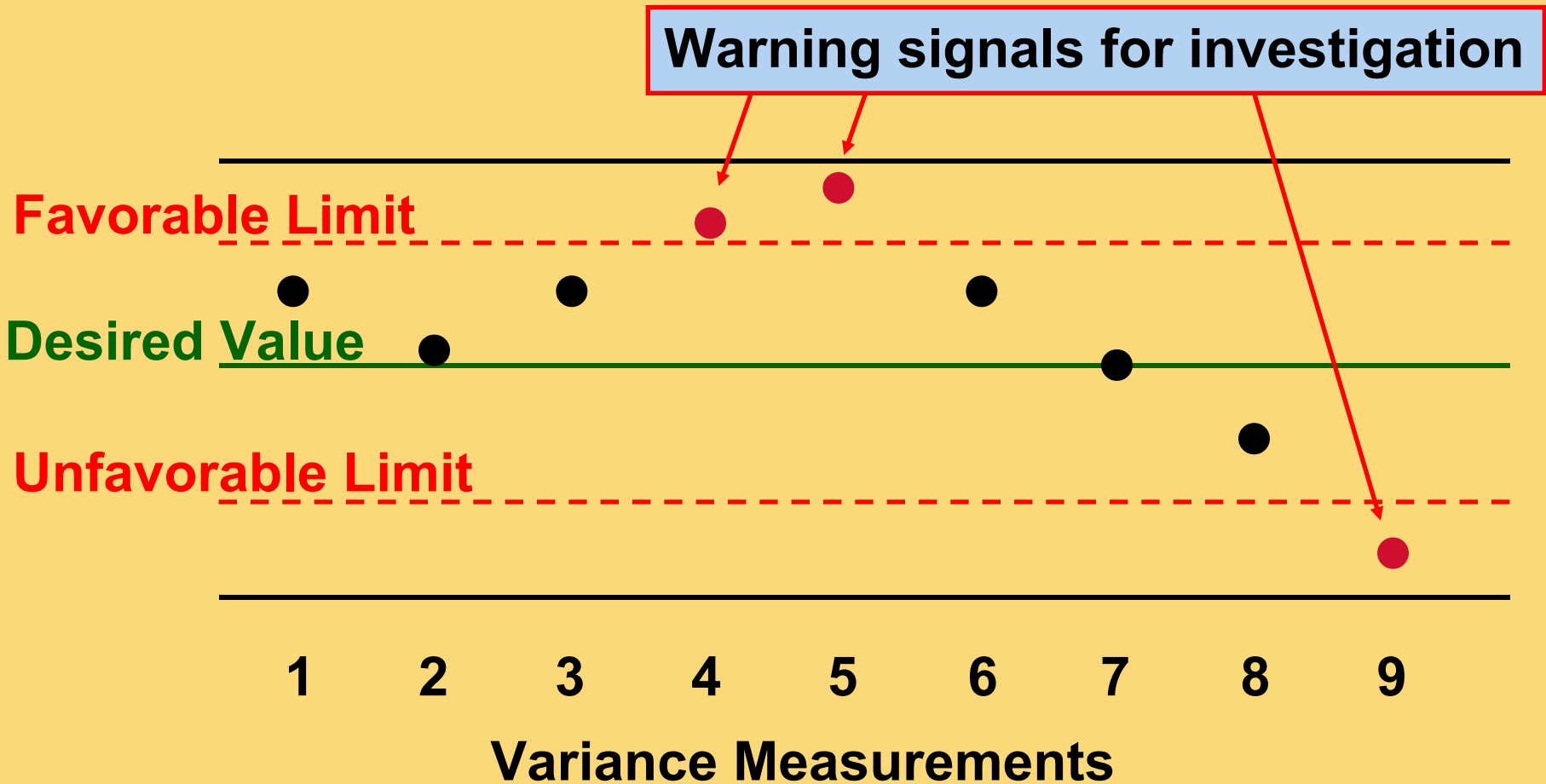


**How do I know
which variances to
investigate?**



**Larger variances, in
dollar amount or as
a percentage of the
standard, are
investigated first.**

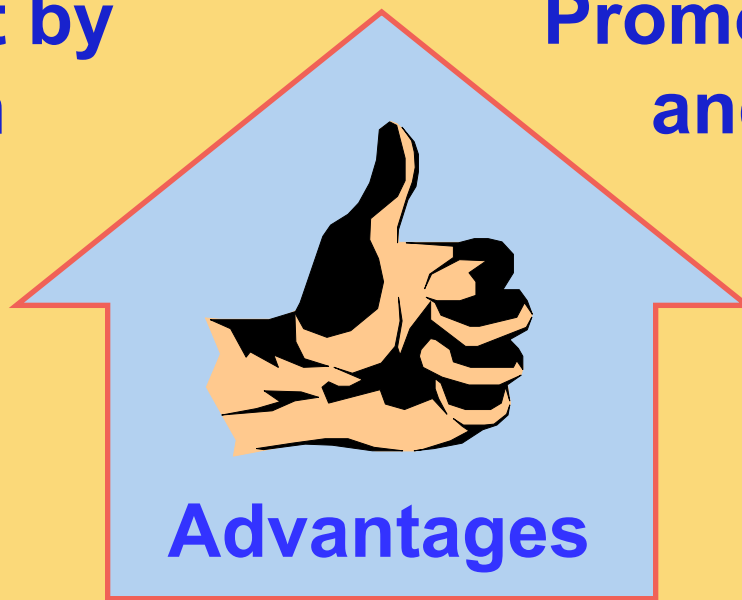
A Statistical Control Chart



Advantages of Standard Costs

**Management by
exception**

**Promotes economy
and efficiency**



**Simplified
bookkeeping**

**Enhances
responsibility
accounting**

Responsibility Accounting

Managers should be held responsible for those items — and *only* those items — that the manager can actually control to a significant extent.

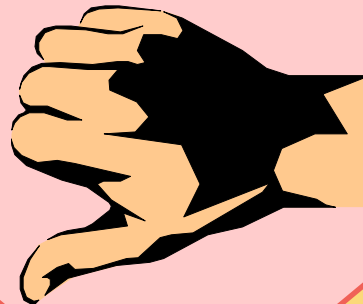


Potential Problems with Standard Costs

Emphasizing standards
may exclude other
important objectives.

Favorable
variances may
be misinterpreted.

Potential Problems



Standard cost
reports may
not be timely.

Emphasis on
negative may
impact morale.

Invalid assumptions
about the relationship
between labor
cost and output.

Continuous
improvement may
be more important
than meeting standards.

- **Standard cost variance reports are usually prepared on a monthly basis and are often released days or weeks after the end of the month; hence, the information can be outdated.**
- **If variances are misused as a club to negatively reinforce employees, morale may suffer and employees may make dysfunctional decisions.**
- **Labor variances make two important assumptions. First, they assume that the production process is labor-paced; if labor works faster, output will go up. Second, the computations assume that labor is a variable cost. These assumptions are often invalid in today's automated manufacturing environment where employees are essentially a fixed cost.**

- In some cases, a “favorable” variance can be as bad or worse than an unfavorable variance.
- Excessive emphasis on meeting the standards may overshadow other important objectives such as maintaining and improving quality, on-time delivery, and customer satisfaction.
- Just meeting standards may not be sufficient; continual improvement may be necessary to survive in a competitive environment.

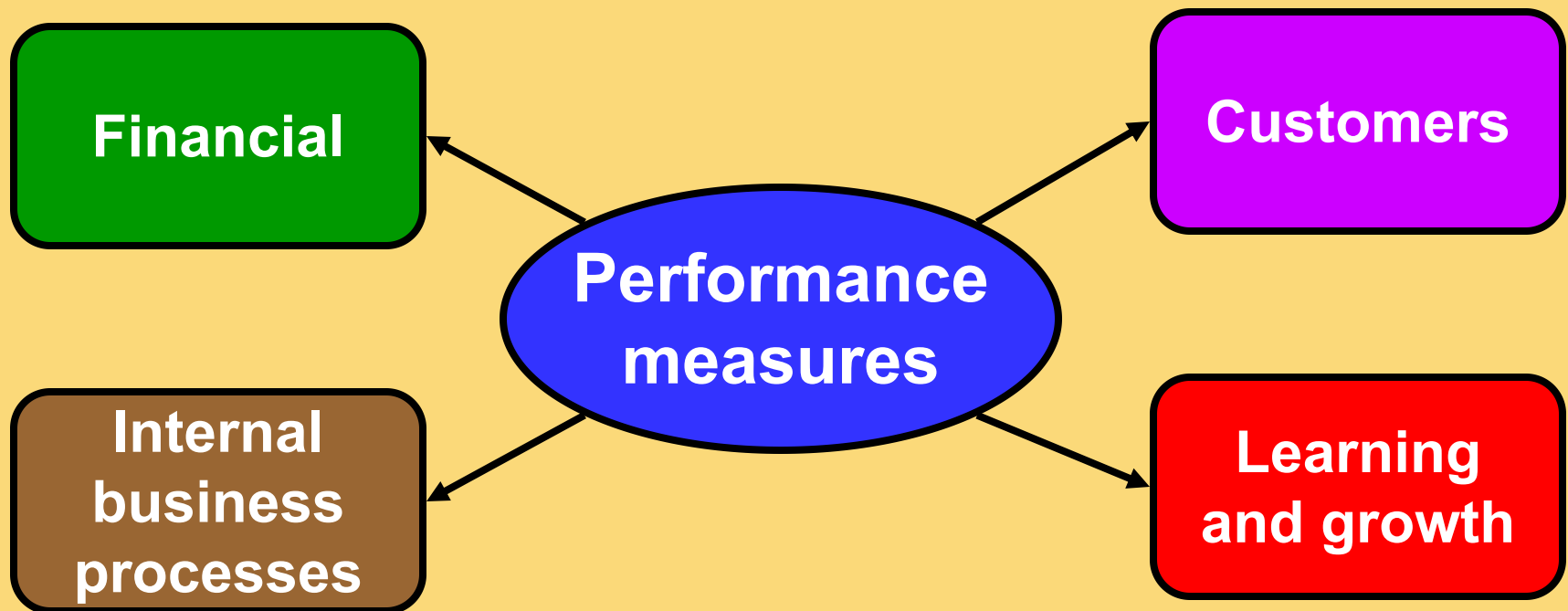
Learning Objective 5



Understand how a
balanced scorecard
fits together and
how it supports a
company's strategy.

The Balanced Scorecard

Management translates its strategy into performance measures that employees understand and accept.



The Balanced Scorecard: From Strategy to Performance Measures

Performance Measures

Financial

Has our financial performance improved?

What are our financial goals?

Customer

Do customers recognize that we are delivering more value?

What customers do we want to serve and how are we going to win and retain them?

Internal Business Processes

Have we improved key business processes so that we can deliver more value to customers?

What internal business processes are critical to providing value to customers?

Vision
and
Strategy

Learning and Growth

Are we maintaining our ability to change and improve?

The Balanced Scorecard: Non-financial Measures

The balanced scorecard relies on non-financial measures in addition to financial measures for two reasons:

❶ Financial measures are lag indicators that summarize the results of past actions. Non-financial measures are leading indicators of future financial performance.

❷ Top managers are ordinarily responsible for financial performance measures – not lower level managers. Non-financial measures are more likely to be understood and controlled by lower level managers.

The Balanced Scorecard for Individuals

The entire organization should have an overall balanced scorecard.

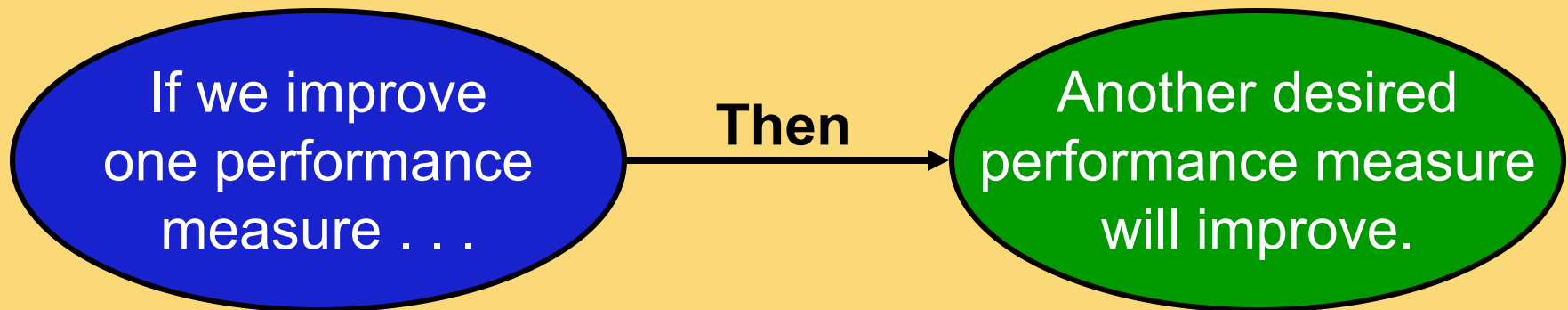
Each individual should have a personal balanced scorecard.



A personal scorecard should contain measures that can be influenced by the individual being evaluated and that support the measures in the overall balanced scorecard.

The Balanced Scorecard

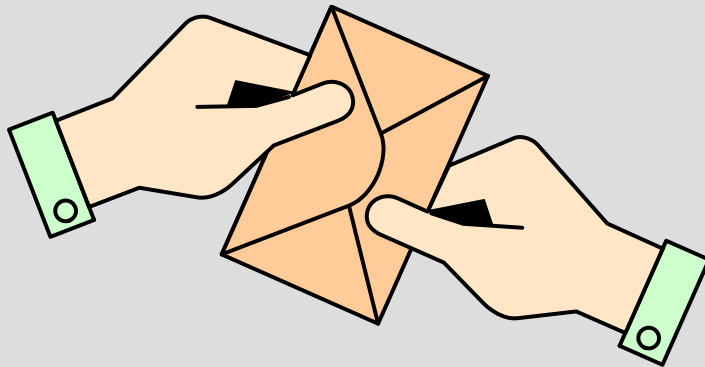
A balanced scorecard should have measures that are linked together on a cause-and-effect basis.



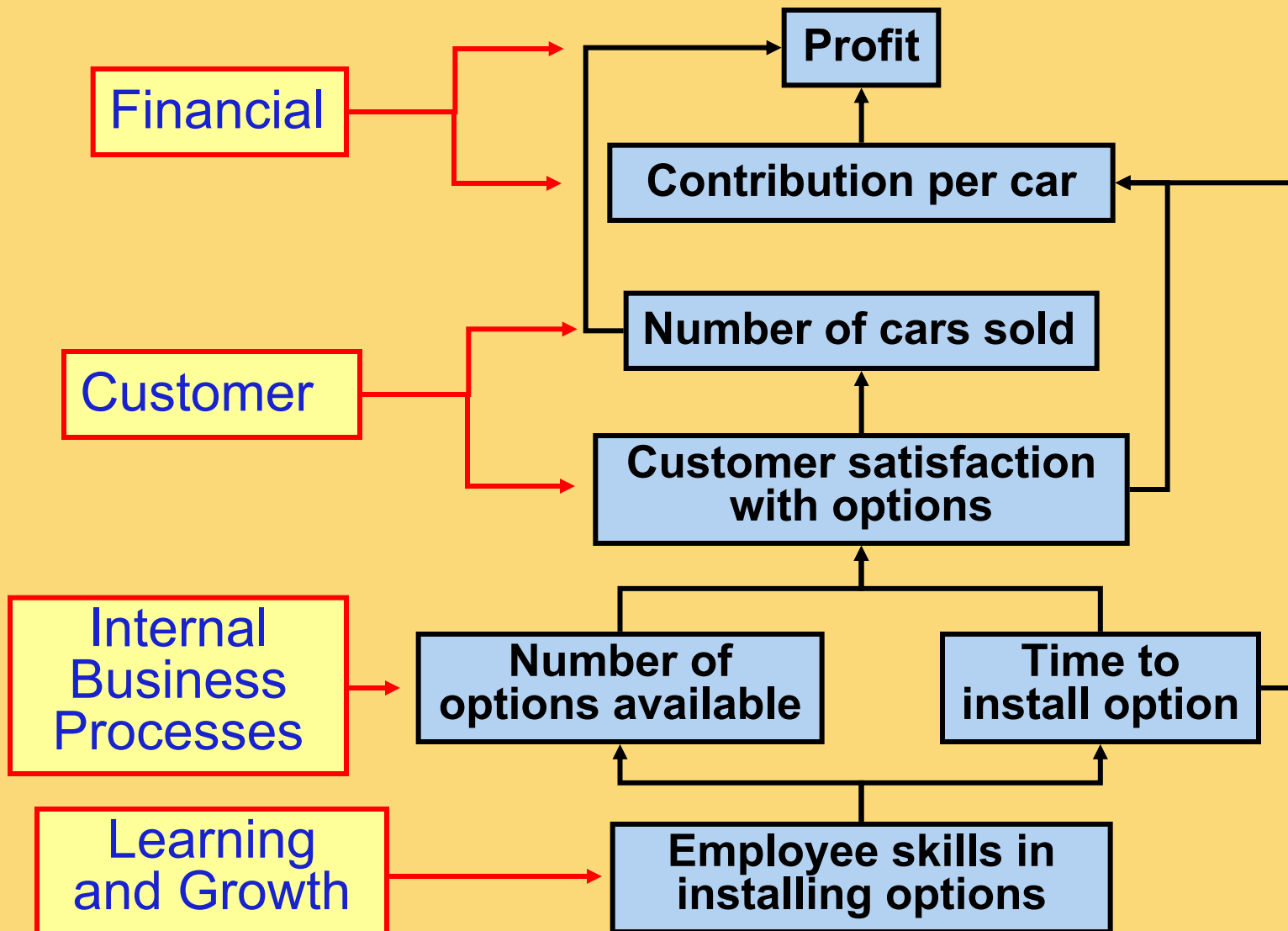
The balanced scorecard lays out concrete actions to attain desired outcomes.

The Balanced Scorecard and Compensation

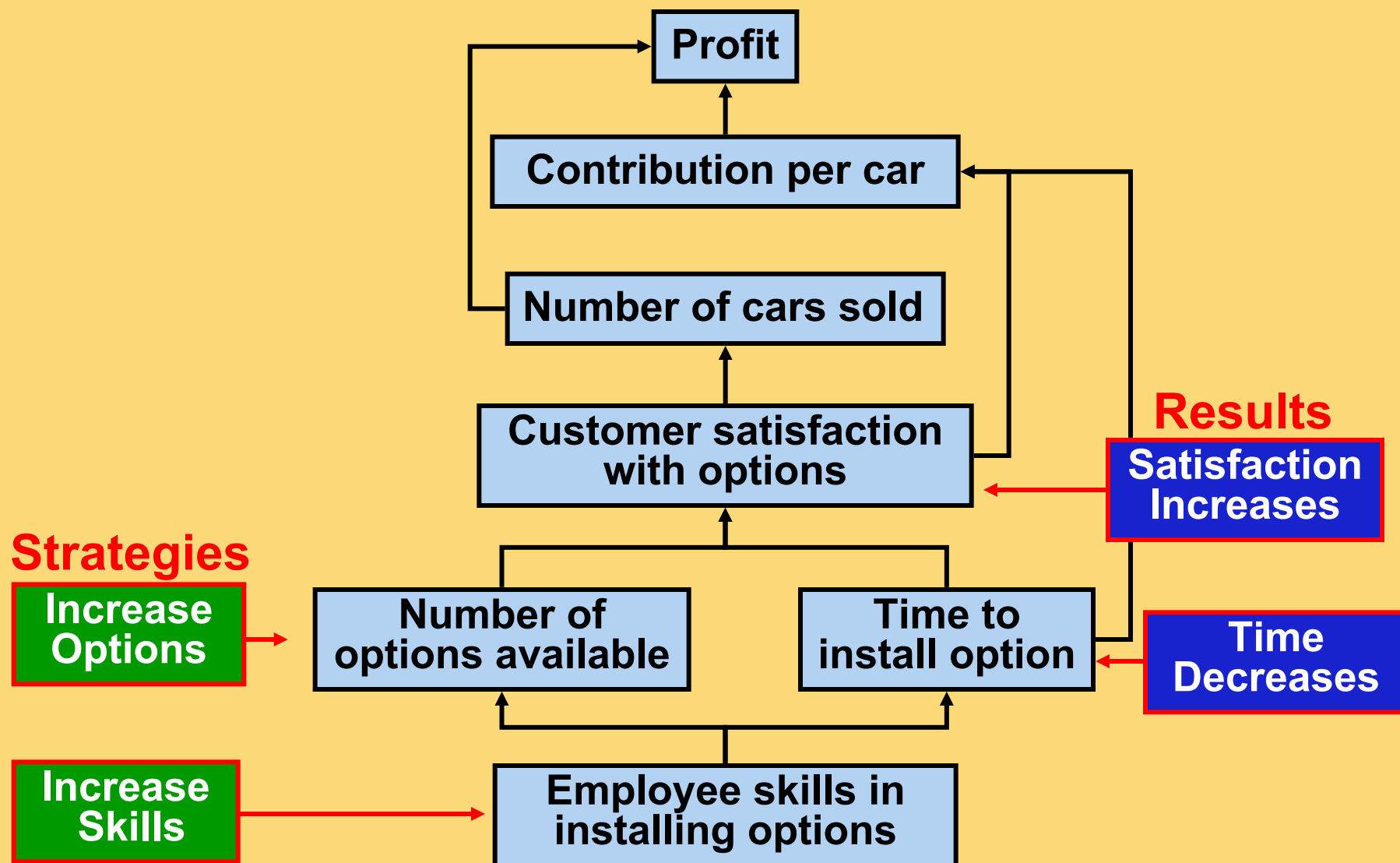
Incentive compensation should be linked to balanced scorecard performance measures.



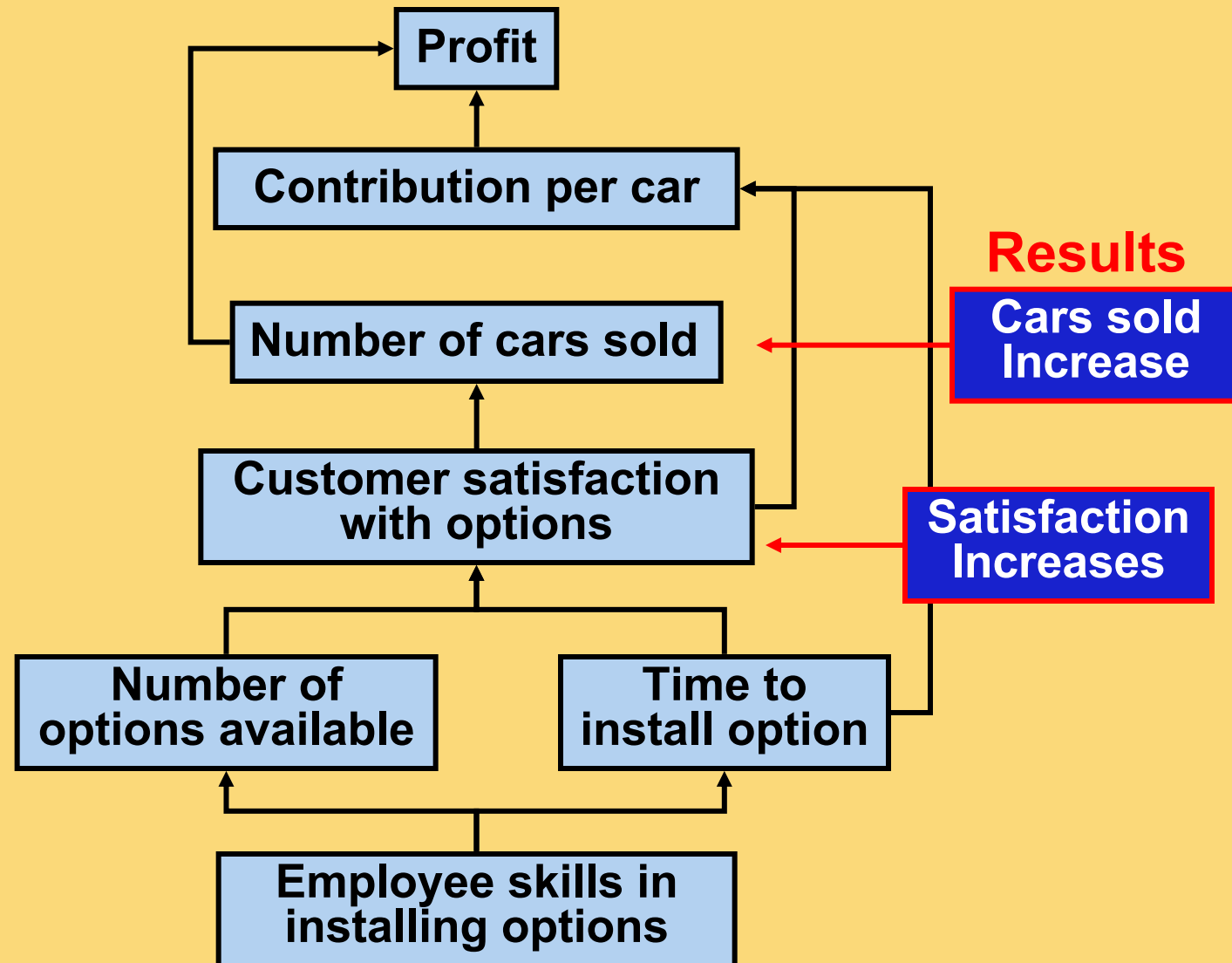
The Balanced Scorecard Jaguar Example



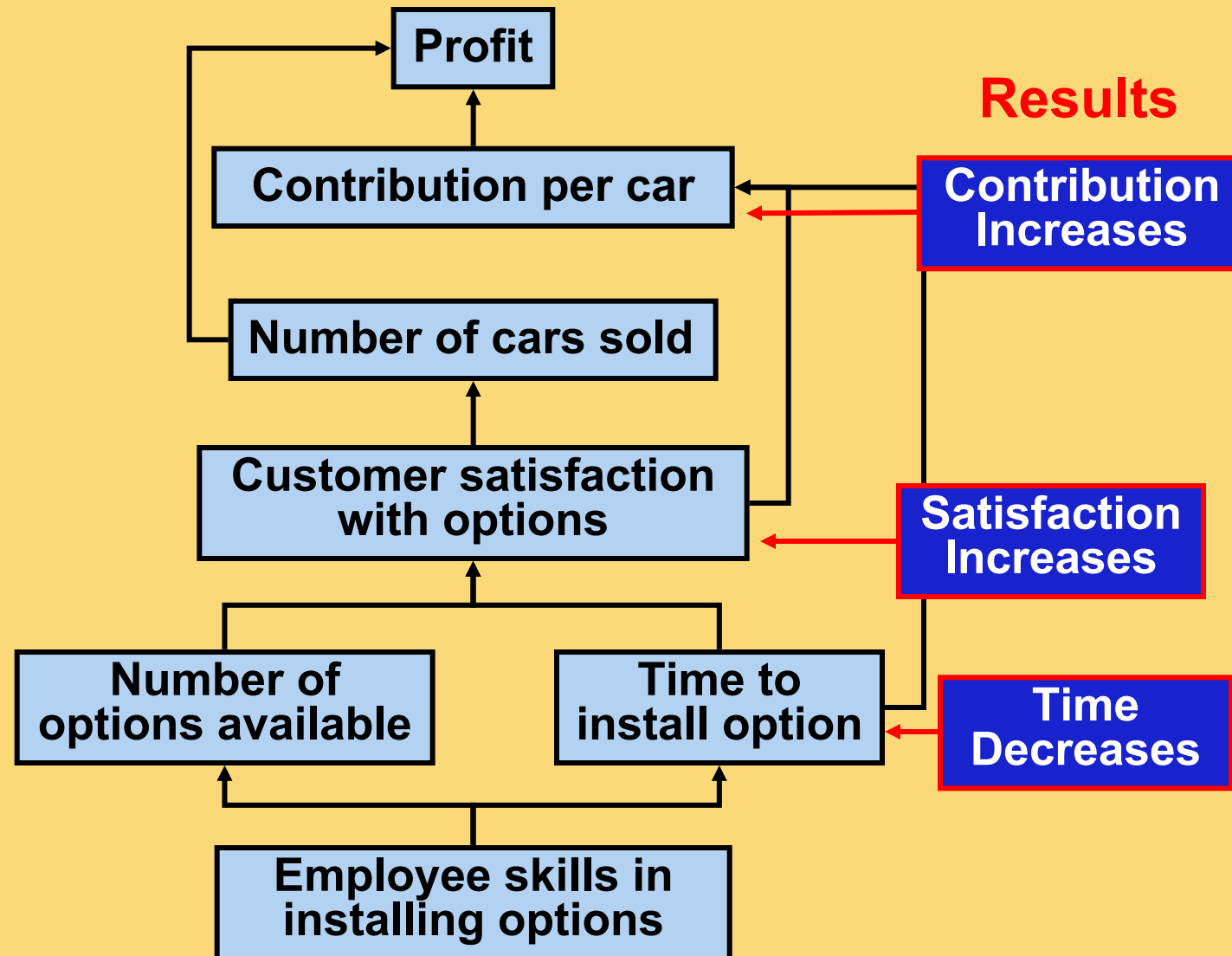
The Balanced Scorecard Jaguar Example



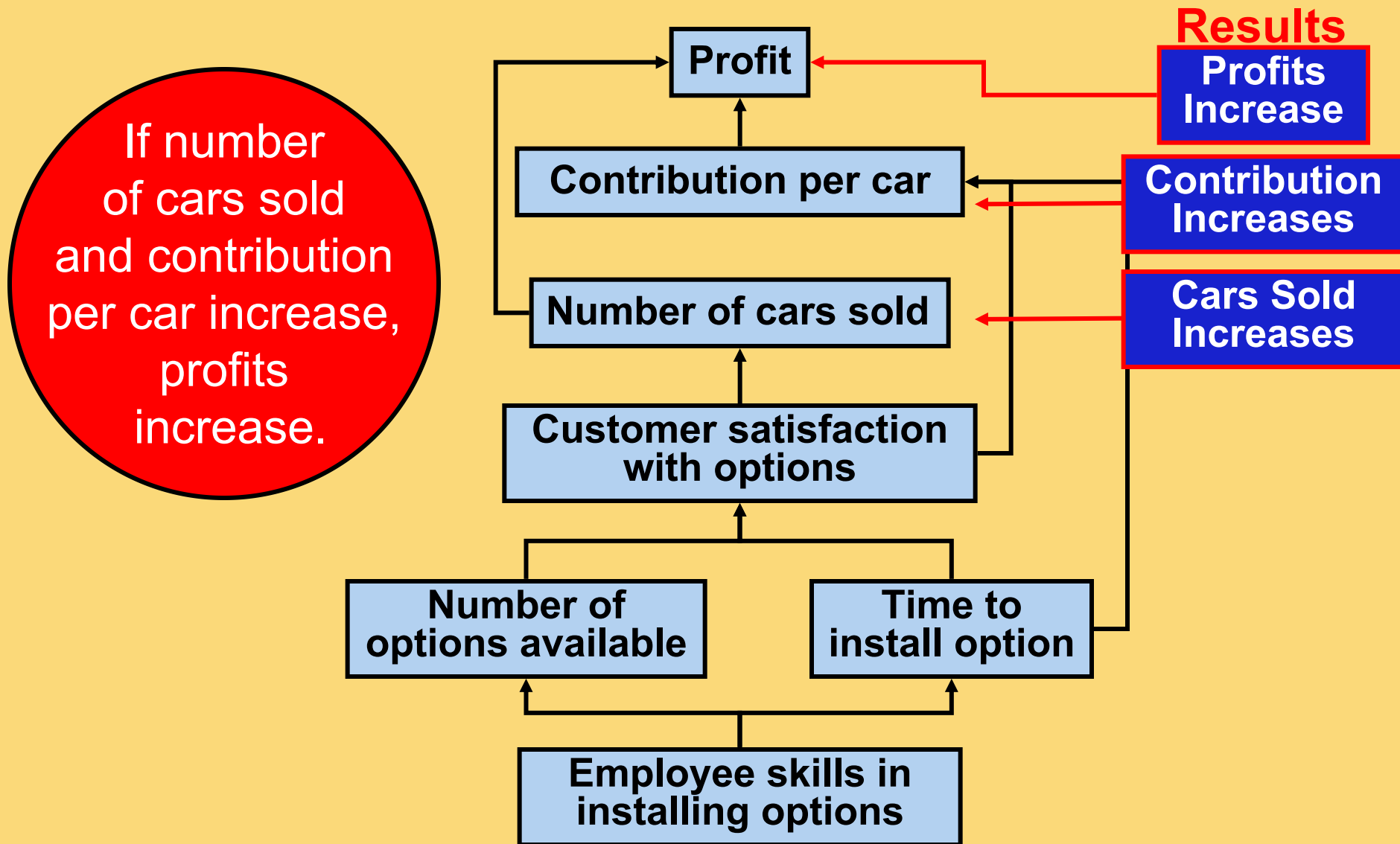
The Balanced Scorecard Jaguar Example



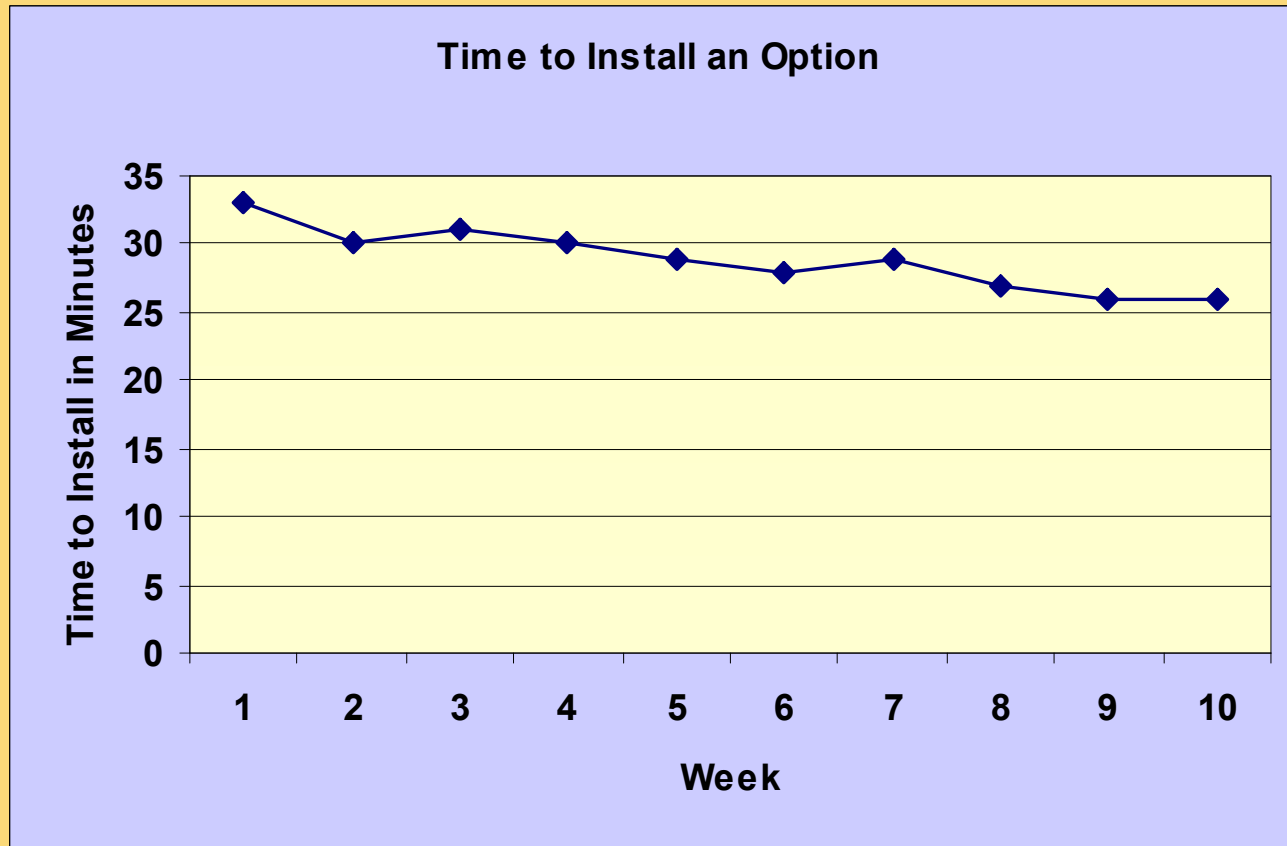
The Balanced Scorecard Jaguar Example



The Balanced Scorecard Jaguar Example

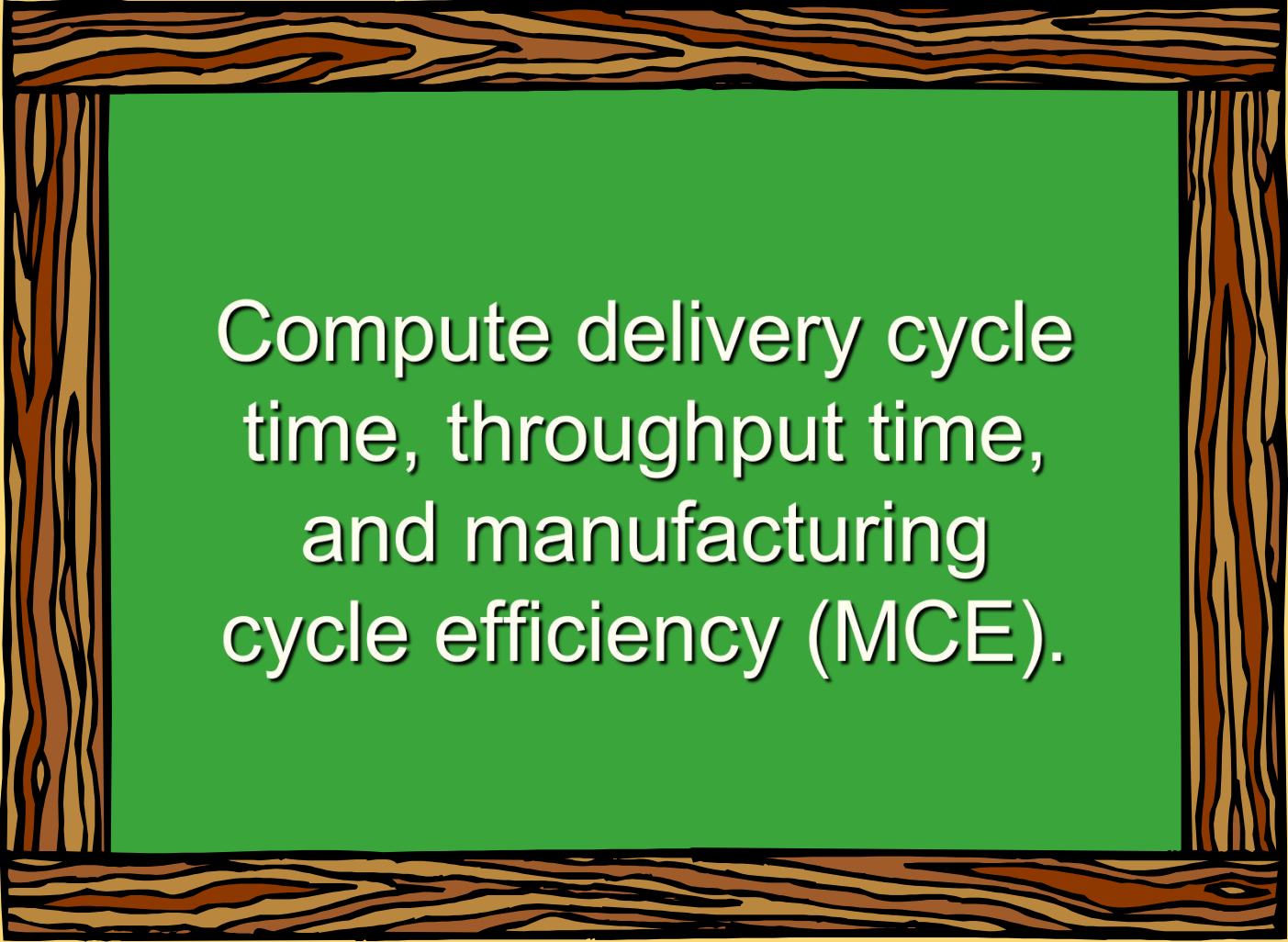


Advantages of Graphic Feedback



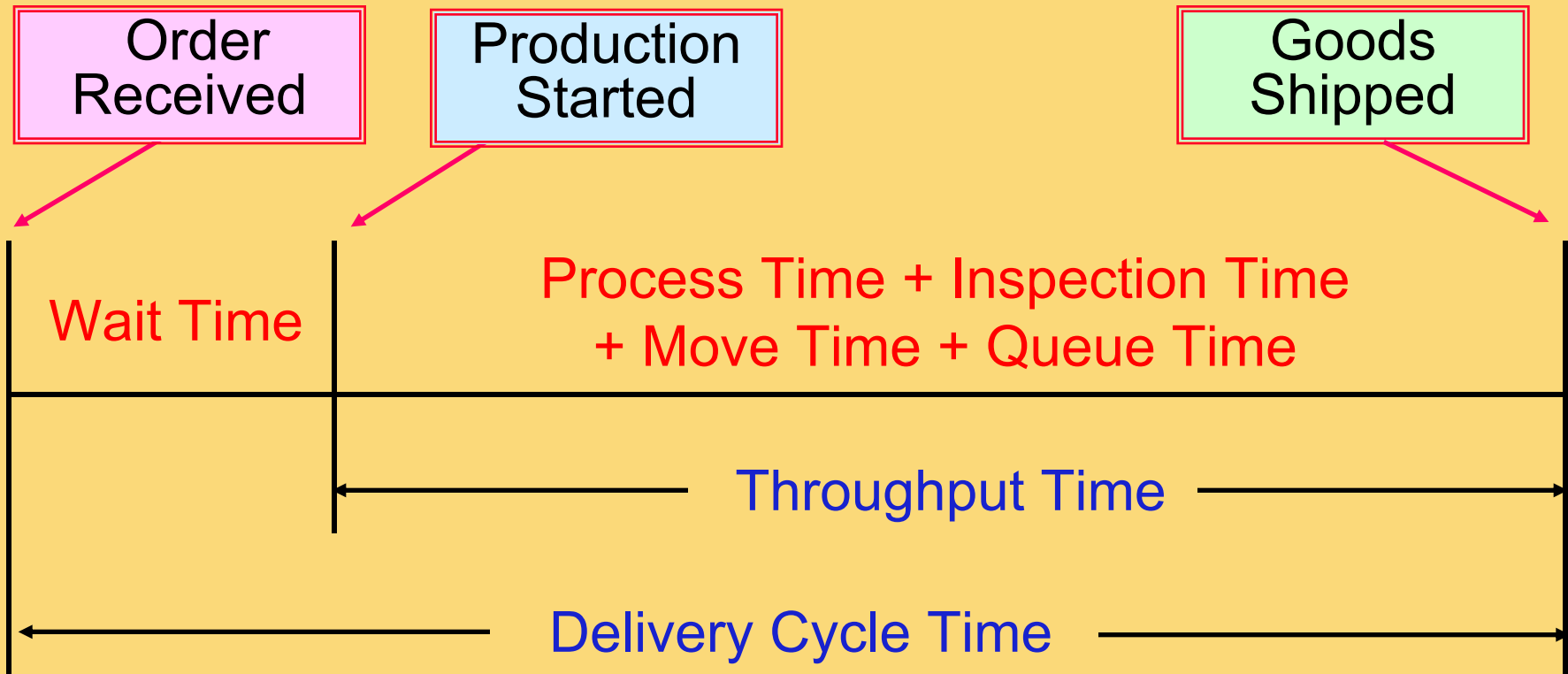
When interpreting its performance, Jaguar will look for continual improvement. It is easier to spot trends or unusual performance if these data are presented graphically.

Learning Objective 6



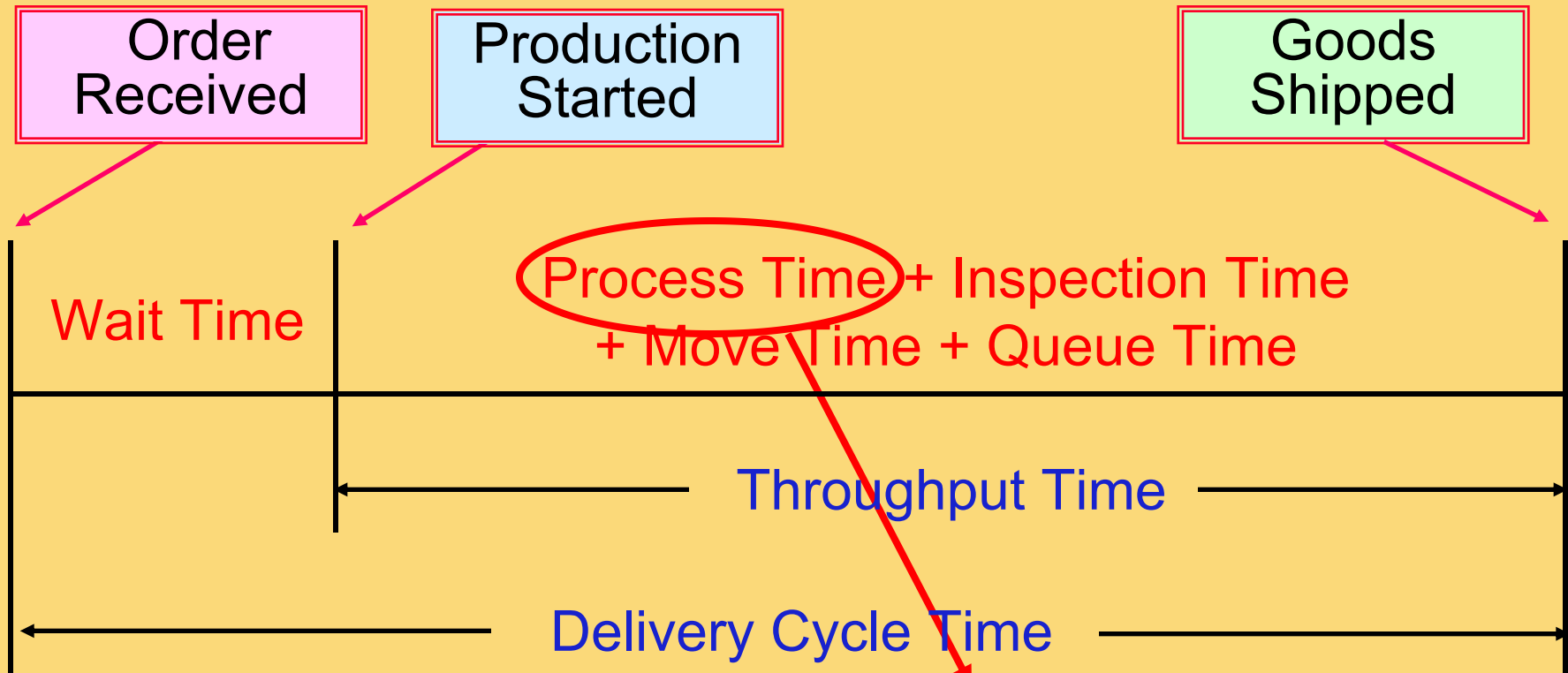
Compute delivery cycle time, throughput time, and manufacturing cycle efficiency (MCE).

Delivery Performance Measures



Process time is the only value-added time.

Delivery Performance Measures



$$\text{Manufacturing Cycle Efficiency} = \frac{\text{Value-added time}}{\text{Manufacturing cycle time}}$$

Quick Check ✓

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the throughput time?

- a. 10.4 days
- b. 0.2 days
- c. 4.1 days
- d. 13.4 days

Quick Check ✓

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the throughput time?

a. 10.4 days

$$\begin{aligned}\text{Throughput time} &= \text{Process} + \text{Inspection} + \text{Move} + \text{Queue} \\ &= 0.2 \text{ days} + 0.4 \text{ days} + 0.5 \text{ days} + 9.3 \text{ days} \\ &= 10.4 \text{ days}\end{aligned}$$

Quick Check ✓

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the Manufacturing Cycle Efficiency?

- a. 50.0%
- b. 1.9%
- c. 52.0%
- d. 5.1%

Quick Check ✓

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the Manufacturing Cycle Efficiency?

a. 50.0%

☒ b. 1.9%

c. 52.0%

d. 5.1%

$$\begin{aligned}\text{MCE} &= \text{Value-added time} \div \text{Throughput time} \\ &= \text{Process time} \div \text{Throughput time} \\ &= 0.2 \text{ days} \div 10.4 \text{ days} \\ &= 1.9\%\end{aligned}$$

Quick Check ✓

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the delivery cycle time?

- a. 0.5 days
- b. 0.7 days
- c. 13.4 days
- d. 10.4 days

$$\begin{aligned}\text{Delivery cycle time} &= \text{Wait time} + \text{Throughput time} \\ &= 3.0 \text{ days} + 10.4 \text{ days} \\ &= 13.4 \text{ days}\end{aligned}$$

A TQM team at Narton Corp has recorded the following average times for production:

Wait	3.0 days	Move	0.5 days
Inspection	0.4 days	Queue	9.3 days
Process	0.2 days		

What is the delivery cycle time?

a. 0.5 days

b. 0.7 days

c. 13.4 days

d. 10.4 days

End of Chapter 10

